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Rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson's disease (Review)

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[Intervention Review]

Rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson's disease

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ABSTRACT

Rationale

Oropharyngeal dysphagia is a common and disabling symptom in people with Parkinson’s disease, affecting the safety and efficiency of swallowing and increasing the risk of malnutrition and aspiration pneumonia. Despite its clinical relevance, the effectiveness of rehabilitation interventions for managing dysphagia in this population remains uncertain.

Objectives

To assess the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in improving swallowing safety and efficiency in people with Parkinson’s disease.

Search methods

We searched the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, Cumulative Index to Nursing and Allied Health Literature, Web of Science, and Speech Pathology Database for Best Interventions and Treatment Efficacy up to 26 September 2025. We also searched ClinicalTrials.gov and the World Health Organization International Clinical Trials Registry Platform to retrieve ongoing and recently completed trials, and OpenGrey for grey literature.

Eligibility criteria

We included randomised controlled trials (RCTs) of rehabilitation interventions aimed at improving oropharyngeal dysphagia in people with Parkinson’s disease. We included any rehabilitation intervention for oropharyngeal dysphagia, alone or in combination with another rehabilitation intervention, with or without concurrent pharmacological treatments. Comparators included no intervention, usual care, a placebo/sham stimulation or attention control intervention, or any other rehabilitation intervention.

Outcomes

Critical outcomes: swallowing safety and swallowing efficiency

Important outcomes: dysphagia severity, saliva management, quality of life, respiratory outcomes, nutritional and dietary outcomes, and adverse events

Risk of bias

We assessed the risk of bias in the included studies using Cochrane's revised risk of bias (RoB 2) tool.

Synthesis methods

When two or more studies were pooled, we conducted a random-effects meta-analysis to calculate standardised mean differences (SMDs) and 95% confidence intervals (CIs). When only one study was available, we used a fixed-effect model to estimate the effect, calculating mean differences (MDs) or odds ratios (ORs) with 95% CIs for all outcomes. We used the GRADE approach to assess the certainty of the evidence.

Included studies

We included 18 studies (1216 participants). Five studies were conducted in Europe, four in the USA, and nine in the rest of the world. Generally, studies were heterogeneous and often poorly reported. We included only two studies in the meta-analysis. Ten studies tested behavioural interventions, while eight studies tested one or more stimulation interventions: neuromuscular treatments (three studies), repetitive transcranial magnetic stimulation (two studies), deep brain stimulation (two studies), and transcranial direct current stimulation (one study).

Synthesis of results

Compared to sham expiratory muscle strength training (EMST), EMST may improve swallowing safety at the end of four weeks of treatment (SMD -0.66, 95% CI -1.04 to -0.28; I^2 = 0%; 2 studies; 113 participants; very low-certainty evidence), but the evidence is very uncertain. Similarly, the evidence is very uncertain about the effect of EMST compared to sham EMST on all the following outcomes: swallowing safety after three months of follow-up (MD -0.18, 95% CI -0.69 to 0.33; 1 study, 45 participants); swallowing efficiency at the end of treatment (MD -4.4, 95% CI -7.5 to -1.3; 1 study, 45 participants), and after three months (MD -1.7, 95% CI -4.58 to 1.18; 1 study, 45 participants); dysphagia severity at the end of treatment (MD -2.78, 95% CI -18.04 to 12.48; 1 study; 45 participants) and after three months (MD -2.9, 95% CI -16.94 to 11.14; 1 study; 45 participants); quality of life at the end of treatment (MD 14.69, 95% CI -63.52 to 92.90; 1 study, 45 participants) and after three months (MD 11.4, 95% CI -67.95 to 90.75; 1 study, 45 participants). Lastly, EMST may result in little to no difference in peak expiratory flow after five weeks of treatment (MD -0.5 L/second, 95% CI -1.05 to 0.05; 1 study, 58 participants; low-certainty evidence).

The evidence is very uncertain about the effect of video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) compared to SLT alone on all the following outcomes: quality of life at the end of treatment (MD 1.62, 95% CI 0.33 to 2.91; 1 study, 42 participants), and after six months (MD 1.86, 95% CI 0.82 to 2.9; 1 study, 42 participants); and dysphagia severity at the end of treatment (MD -0.7, 95% CI -5.13 to 3.73; 1 study, 42 participants) and in the long term (MD -4.03, 95% CI -7.85 to -0.21; 1 study, 42 participants).

The evidence is very uncertain about the effect of neuromuscular electrical stimulation (NMES) plus SLT compared to sham NMES plus SLT on swallowing safety (MD -1.34, 95% CI -2.32 to -0.36; 1 study, 18 participants; very low-certainty evidence), and swallowing efficiency at the end of four weeks of treatment (MD -2.34, 95% CI -7.16 to 2.48; 1 study, 18 participants; very low-certainty evidence).

The evidence is also very uncertain about the effect of tongue strength training plus SLT compared to SLT alone on dysphagia severity after eight weeks of treatment (MD 0.13, 95% CI -0.17 to 0.43; 1 study, 60 participants; very low-certainty evidence).

Our certainty in these findings is very low due to imprecision (small sample size), high risk of bias related to deviations from intended interventions, measurement of the outcome, and the randomisation process, and poor reporting.

Authors' conclusions

Although the number of RCTs has increased since the publication of a previous Cochrane review on the topic in 2001, uncertainty remains regarding the effectiveness of swallowing rehabilitation in individuals with Parkinson’s disease. Behavioural interventions, particularly EMST, may improve swallowing safety by reducing penetration and aspiration, but the evidence is very uncertain. Most included studies were small and methodologically limited, constraining the robustness of conclusions. Large, well-designed, placebo-controlled trials are required to assess the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson’s disease.

Funding

No funding was received for this work.

Registration

Protocol (2024) DOI: 10.1002/14651858.CD015816

PLAIN LANGUAGE SUMMARY

Do rehabilitation treatments help people with Parkinson's disease and swallowing disorders?

Key messages

- Many people who have Parkinson's disease develop swallowing disorders, known as 'oropharyngeal dysphagia' or just 'dysphagia'. This condition can cause serious health problems, increase the risk of death, and reduce people's quality of life.
- We do not know if various types of rehabilitation for dysphagia help people with Parkinson's disease to swallow more effectively and safely because the evidence is limited and of poor quality.
- We need larger, high-quality studies to help us understand which, if any, rehabilitation treatments work best.

What is oropharyngeal dysphagia?

Oropharyngeal dysphagia (hereafter referred to as 'dysphagia') is a swallowing problem in which it is difficult or unsafe to move food or drink from the mouth through the throat and into the stomach.

Up to 80% of people with Parkinson's disease develop dysphagia because the disease affects brain regions that control movement, weakening and slowing the muscles involved in swallowing. As a result, food or drink may not move smoothly into the stomach and can sometimes enter the airway instead of the oesophagus (food pipe).

Dysphagia can lead to serious health problems, including malnutrition and aspiration pneumonia (when food or drink is inhaled into the airway or lungs), increase the risk of death, and reduce quality of life.

How is dysphagia treated in people with Parkinson's disease?

In people with Parkinson's disease, dysphagia is usually treated first with medicines that aim to improve the symptoms of Parkinson's disease overall. If swallowing problems continue, rehabilitation treatments may be used to try to make swallowing safer and easier. These include:

- exercises to strengthen the breathing or swallowing muscles;
- changes to head or body position during eating and drinking;
- changes to food and drink textures;
- stimulation treatments, which use safe, gentle medical devices to deliver electrical or magnetic stimulation to the brain or the nerves and muscles involved in swallowing, with the aim of helping activate and improve swallowing.

What did we want to find out?

We wanted to know if rehabilitation interventions for oropharyngeal dysphagia (swallowing disorders) help people with Parkinson's disease to swallow food and drink more effectively and safely.

What did we do?

We searched for studies that compared any type of rehabilitation treatment (as a single treatment or combined with another form of rehabilitation or medicine) to:

- no treatment;
- usual care (standard care that people with Parkinson's disease and dysphagia receive);
- 'sham' rehabilitation, where the treatment is fake;
- an inactive treatment, known as a 'placebo';
- another type of dysphagia rehabilitation.

We limited our focus to adults (18 years and older) with Parkinson's disease. We were interested in the effects of treatment on:

- swallowing efficiency and safety;
- dysphagia severity;
- quality of life;
- breathing issues;
- unwanted or harmful events (known as 'adverse events').

We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We found 18 studies involving 1216 people with Parkinson's disease. Four studies were conducted in the USA, five in Europe, and nine in the rest of the world. Participants' average age ranged from 58.8 to 76.2 years. Most participants were male. All studies compared rehabilitation treatments to usual care or sham/placebo treatments. Treatment durations varied widely, from as short as two weeks to as long as one year. The studies were conducted in hospitals and people's homes.

We classified the studies into two broad groups: (1) behavioural treatments and (2) stimulation treatments.

Ten studies explored behavioural treatments, such as expiratory muscle strength training (i.e. exercises aimed at strengthening the muscles we use to breathe out) and tongue strength training (i.e. exercises aimed at strengthening the tongue muscles).

Eight studies investigated one or more of a range of stimulation treatments, as follows:

- neuromuscular electrical stimulation, where a low-intensity electrical current activates the muscles (four studies);
- repetitive transcranial magnetic stimulation, where magnetic pulses are applied to the scalp to stimulate areas of the brain involved in swallowing (two studies);
- deep brain stimulation, where electrical signals are delivered to specific areas deep in the brain to help control movement problems in Parkinson's disease (two studies);
- transcranial direct current stimulation, where a very weak electrical current is applied to the scalp to change brain activity (one study).

All studies assessed outcomes immediately after the treatment. Six studies also assessed outcomes at anywhere from one to 14 months after treatment.

Only twelve studies provided data suitable for analysis. The evidence was very uncertain about the effects of all the different rehabilitation treatments on our key outcomes of interest (i.e. swallowing efficiency and safety, dysphagia severity, quality of life, and breathing issues).

There was very limited information about unwanted or harmful events.

What are the limitations of the evidence?

The studies were generally small and of poor quality. Also, not all the studies provided information about participants, treatments, and outcomes. Consequently, we have very low confidence in the evidence.

How current is this evidence?

The evidence is current to 26 September 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Expiratory muscle strength training (EMST) compared to sham EMST for oropharyngeal dysphagia in people with Parkinson's disease

Expiratory muscle strength training (EMST) compared to sham EMST for oropharyngeal dysphagia in people with Parkinson's disease

Patient or population: oropharyngeal dysphagia in people with Parkinson's disease
Setting: inpatient and oupatient
Intervention: expiratory muscle strength training (EMST)
Comparison: sham EMST

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with sham EMST	Risk with expiratory muscle strength training (EMST)				
Swallowing efficiency assessed with: Bolus Residue Scale follow-up: mean 4 weeks	The mean swallowing efficiency was **10.57** score	MD **4.4 score lower** (7.5 lower to 1.3 lower)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on swallowing efficiency measured with the Bolus Residue Scale at the end of treatment.
Swallowing efficiency assessed with: Bolus Residue Scale follow-up: mean 3 months	The mean swallowing efficiency was **9.95** score	MD **1.7 score lower** (4.58 lower to 1.18 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on swallowing efficiency measured with the Bolus Residue Scale at the follow-up.
Swallowing safety assessed with: Penetration-Aspiration Scale (PAS) follow-up: mean 4 weeks	-	SMD **0.66 lower** (1.04 lower to 0.28 lower)	-	113 (2 RCTs)	⊕⊝⊝⊝ Very low[a,c]	The evidence is very uncertain about the effect of EMST on swallowing safety measured with PAS at the end of treatment.
Swallowing safety assessed with: PAS follow-up: mean 3 months	The mean swallowing safety was **0.43** score	MD **0.18 score lower** (0.69 lower to 0.33 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on swallowing safety measured with PAS at the follow-up.
Dysphagia severity assessed with: Swallowing Disturbance Questionnaire (SDQ) follow-up: mean 4 weeks	The mean dysphagia severity was **9.36** score	MD **2.78 score lower** (18.04 lower to 12.48 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on dysphagia severity measured with SDQ at the end of treatment.

Dysphagia severity assessed with: SDQ follow-up: mean 3 months	The mean dysphagia severity was **10.07** score	MD **2.9 score lower** (16.94 lower to 11.14 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on dysphagia severity measured with SDQ at the follow-up.
Saliva management - not measured	-	-	-	-	-	
Quality of life assessed with: Swallowing Quality of Life (SWAL-QoL) - total score follow-up: mean 4 weeks	The mean quality of life was **174.19** score	MD **14.69 score higher** (63.52 lower to 92.9 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on quality of life measured with SWAL-QoL (total score) at the end of treatment.
Quality of life assessed with: SWAL-QoL - total score follow-up: mean 3 months	The mean quality of life was **174.52** score	MD **11.4 score higher** (67.95 lower to 90.75 higher)	-	45 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of EMST on quality of life measured with SWAL-QoL (total score) at the follow-up.
Peak expiratory flow assessed with: L/second follow-up: mean 5 weeks	The mean peak expiratory flow was **3.60** L/sec	MD **0.5 L/sec lower** (1.05 lower to 0.05 higher)	-	58 (1 RCT)	⊕⊕⊝⊝ Low[d]	The evidence suggests that EMST results in little to no difference in peak expiratory flow.
Adverse events - not measured	-	-	-	-	-	

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **SMD:** standardised mean difference

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457183911737941556.

[a] Dowgraded one level for the high risk of bias related to deviations from intended interventions
[b] Downgraded two levels for imprecision (fewer than 100 participants and wide CI)
[c] Downgraded two levels for imprecision (CI crossed two threshold levels)
[d] Downgraded two levels for imprecision (fewer than 100 participants)

Summary of findings 2. Summary of findings table - Video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) compared to speech-language therapy (SLT) for oropharyngeal dysphagia in people with Parkinson's disease

Video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) compared to speech-language therapy (SLT) for oropharyngeal dysphagia in people with Parkinson's disease

Patient or population: oropharyngeal dysphagia in people with Parkinson's disease
Setting: inpatient
Intervention: video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT)
Comparison: speech-language therapy (SLT)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with speech-language therapy (SLT)	Risk with video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT)				
Swallowing efficiency - not measured	-	-	-	-	-	
Swallowing safety - not measured	-	-	-	-	-	
Dysphagia severity assessed with: Swallowing Disturbance Questionnaire (SDQ) follow-up: mean 4 weeks	The mean dysphagia severity was **13.43** score	MD **0.7 score lower** (5.13 lower to 3.73 higher)	-	42 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of VAST plus SLT on dysphagia severity at the end of treatment.
Dysphagia severity assessed with: SDQ follow-up: mean 6 months	The mean dysphagia severity was **13.08** score	MD **4.03 score lower** (7.85 lower to 0.21 lower)	-	42 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of VAST plus SLT on dysphagia severity at the follow-up.
Saliva management - not measured	-	-	-	-	-	
Quality of life assessed with: Swallowing Quality of Life (SWAL-QoL) - Burden impact follow-up: mean 4 weeks	The mean quality of life was **7.26** score	MD **1.62 score higher** (0.33 higher to 2.91 higher)	-	42 (1 RCT)	⊕⊝⊝⊝ Very low[a,c]	The evidence is very uncertain about the effect of VAST plus SLT on quality of life (burden impact) at the end of treatment.

Quality of life assessed with: SWAL-QoL - Burden impact follow-up: mean 6 months	The mean quality of life was **6.58** score	MD **1.86 score higher** (0.82 higher to 2.9 higher)	-	42 (1 RCT)	⊕⊝⊝⊝ Very low[a,c]	The evidence is very uncertain about the effect of VAST plus SLT on quality of life (burden impact) at the follow-up.
Respiratory outcomes - not measured	-	-	-	-	-	
Adverse events - not measured	-	-	-	-	-	

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457189006549192721.

[a] Downgraded one level for high risk of bias related to the randomisation process
[b] Downgraded two levels for imprecision (fewer than 100 participants and wide CI)
[c] Downgraded two levels for imprecision (fewer than 100 participants).

Summary of findings 3. Summary of findings table - Neuromuscular electrical stimulation (NMES) plus speech-language therapy (SLT) compared to sham NMES plus SLT for oropharyngeal dysphagia in people with Parkinson's disease

Neuromuscular electrical stimulation (NMES) plus speech-language therapy (SLT) compared to sham NMES plus SLT for oropharyngeal dysphagia in people with Parkinson's disease

Patient or population: oropharyngeal dysphagia in people with Parkinson's disease
Setting: inpatient
Intervention: neuromuscular electrical stimulation (NMES) plus speech-language therapy (SLT)
Comparison: sham NMES plus SLT

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments

	Risk with sham NMES plus SLT	Risk with neuromuscular electrical stimulation (NMES) plus speech-language therapy (SLT)				
Swallowing efficiency assessed with: Videofluoroscopic Dysphagia Scale (VDS) - Pharyngeal phase follow-up: mean 4 weeks	The mean swallowing efficiency was **39.06** score	MD **2.34 score lower** (7.16 lower to 2.48 higher)	-	18 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of neuromuscular electrical stimulation plus speech-language therapy on swallowing efficiency.
Swallowing safety assessed with: Penetration-Aspiration Scale (PAS) - Semi-solid consistency follow-up: mean 4 weeks	The mean swallowing safety was **3.67** score	MD **1.34 score lower** (2.32 lower to 0.36 lower)	-	18 (1 RCT)	⊕⊝⊝⊝ Very low[a,c]	The evidence is very uncertain about the effect of neuromuscular electrical stimulation plus speech-language therapy on swallowing safety.
Dysphagia severity - not measured	-	-	-	-	-	
Saliva management - not measured	-	-	-	-	-	
Quality of life - not measured	-	-	-	-	-	
Respiratory outcomes - not measured	-	-	-	-	-	
Adverse events - not measured	-	-	-	-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457190623634827532.

[a] Downgraded one level for high risk of bias in the measurement of the outcome and some concerns for bias in the randomisation process and selection of the reported results
[b] Downgraded two levels for imprecision (fewer than 100 participants and wide CI)
[c] Downgraded two levels for imprecision (fewer than 100 participants)

Summary of findings 4. Summary of findings table - Tongue strength training plus speech-language therapy (SLT) compared to speech-language therapy (SLT) alone for oropharyngeal dysphagia in people with Parkinson's disease

Tongue strength training plus speech-language therapy (SLT) compared to speech-language therapy (SLT) alone for oropharyngeal dysphagia in people with Parkinson's disease

Patient or population: oropharyngeal dysphagia in people with Parkinson's disease
Setting: inpatient
Intervention: tongue strength training plus speech-language therapy (SLT)
Comparison: speech-language therapy (SLT) alone

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with speech-language therapy (SLT) alone	Risk with tongue strength training plus speech-language therapy (SLT)				
Swallowing efficiency - not measured	-	-	-	-	-	
Swallowing safety - not measured	-	-	-	-	-	
Dysphagia severity assessed with: Functional Oral Intake Scale (FOIS) follow-up: mean 8 weeks	The mean dysphagia severity was **6.00** score	MD **0.13 score higher** (0.17 lower to 0.43 higher)	-	60 (1 RCT)	⊕⊝⊝⊝ Very low[a,b]	The evidence is very uncertain about the effect of tongue strength training plus speech-language therapy on dysphagia severity.
Saliva management - not measured	-	-	-	-	-	
Quality of life - not measured	-	-	-	-	-	
Respiratory outcomes - not measured	-	-	-	-	-	
Adverse events - not measured	-	-	-	-	-	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
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[a] Downgraded one level for high risk of bias in measurement of the outcome
[b] Downgraded two levels for imprecision (fewer than 100 participants and wide CI)

BACKGROUND

Parkinson's disease is estimated to affect 0.3% of people in industrialised countries, reaching 1% in people over 60 years old and 3% in people over 80 years [1, 2, 3]. Swallowing disorders (dysphagia) are highly prevalent and clinically significant symptoms in people with Parkinson's disease [4, 5]. A meta-analysis conducted by Kalf and colleagues reported a dysphagia prevalence of up to 82% in people with Parkinson's disease [6]. Dysphagia in Parkinson's disease can lead to severe consequences: aspiration pneumonia, prolonged hospitalisation, increased mortality rate, and reduced quality of life [7, 8, 9]. Despite the high prevalence of dysphagia and its severe clinical sequelae in Parkinson's disease, there is a lack of scientific evidence on rehabilitation interventions. Since roughly 2013, there has been an increase in studies aiming to maintain or improve swallowing safety and efficiency, reduce associated health complications, and increase swallowing-related quality of life in people with Parkinson's disease [10, 11, 12].

Description of the condition

Dysphagia can be broadly categorised based on the anatomical location of the swallowing impairment. Oropharyngeal dysphagia refers to swallowing difficulty predominantly at the oral (mouth) and pharyngeal (upper portion of the throat known as the pharynx) phases, and it differs from oesophageal dysphagia, which affects the oesophagus. Oropharyngeal dysphagia seriously affects people with Parkinson's disease, and it is recognised to be a fatal symptom, significantly increasing the risk of hospitalisation and death in this population [7, 13, 14].

The clinical signs of oropharyngeal dysphagia differ enormously in people with Parkinson's disease. In the literature, it is widely accepted that major signs of oropharyngeal dysphagia, such as aspiration of food and fluid into the airway, occur in the advanced phase of the disease [15, 16], but symptoms may also occur in the early stages of Parkinson's disease [17]. The neuropathological mechanism of swallowing disorders in Parkinson's disease is not yet clearly identified. Dysphagia seems to be caused by a degeneration of subcortical regions such as the basal ganglia and the swallowing central pattern generator in the medulla, as well as of cortical areas [5, 18, 19]. One of the most severe signs of dysphagia is aspiration of food into the airway, which increases the risk of developing pneumonia [20, 21]. Aspiration pneumonia is a lung infection caused by aspiration of food, fluids, or secretions into the airway [20]. A recent retrospective study reported that people with Parkinson's disease had a 2.23 times higher risk for pneumonia than the general population [22]. Furthermore, the presence of aspiration pneumonia was correlated with the mortality rate. Won and colleagues found that nearly 24% of people with Parkinson's disease died within the first year of aspiration pneumonia occurrence, and 91.8% died five years following the diagnosis of aspiration pneumonia [22]. However, the mechanism behind aspiration is complex. There is emerging evidence that reduced voluntary cough expiratory airflow and reflex cough parameters are significantly correlated with penetration and aspiration of bolus material (i.e. a small, rounded mass of chewed food and secretions) during swallowing, leading to aspiration pneumonia [23, 24, 25, 26, 27]. In other words, there seems to be an association between someone's ability to cough, both voluntarily and involuntarily, and the likelihood that particles of chewed food will be inhaled into the airway, leading to aspiration pneumonia.

In general, dysphagic symptoms are similar to those affecting the limb muscles, including bradykinesia (motor control impairment, slow movements, and 'freezing'), muscle rigidity, and prolonged initiation and reaction time [4]. The oral preparatory phase is particularly compromised in Parkinson's disease, resulting in difficulties in bolus formation, manipulation and control, and prolonged swallowing time. Hypotonia (decreased muscle tone) and bradykinesia of the facial muscles can result in difficulties with mastication (chewing) and increased drooling [4, 28]. Tongue tremors, lingual pumping, freezing and festination (involuntary acceleration) movements impact oral transit time, mainly during swallowing solids rather than swallowing softer consistencies [17, 29, 30, 31]. As a result, the oral phase, which involves the propulsion of a bolus into the pharynx, can be severely compromised and prolonged [30, 32]. Difficulties in the oral phase lead to an increase in pharyngeal bolus residues of both food and medication [5, 33, 34, 35, 36]. Pharyngeal residues could be caused by either pre-swallow spillage into the pharynx or post-swallow residue in the pharynx. A recent study showed the presence of medications in the pharynx in a sample of people with Parkinson's disease, suggesting the effects of dysphagia in reducing medication intake and increasing Parkinsonian symptoms [17, 37]. Masticatory difficulties and difficulties at mealtime are significantly correlated with the risk of malnutrition in this population [38]. When oral food intake is severely compromised, an individual's nutrition can be supported by enteral nutrition methods, such as nasogastric tube or percutaneous endoscopic gastrostomy (PEG).

In addition to the nutritional implications, swallowing impairments can have psychological and social impacts on people with Parkinson's disease. In the literature, it is well documented that dysphagia has a direct impact on social life, decreasing mealtime enjoyment and reducing quality of life [39, 40, 41]. These impacts have consequences not just for people with Parkinson's disease and oropharyngeal dysphagia but also for their families [9, 42, 43].

Oropharyngeal dysphagia also impacts healthcare costs. According to data from two studies, there was a 40% higher expenditure on patients with oropharyngeal dysphagia compared to their non-dysphagic counterparts [44, 45]. In hospitalised people with Parkinson's disease, dysphagia was associated with a 44% longer length of stay (adjusted rate ratio 1.44, 95% confidence interval (CI) 1.43 to 1.45) and 46% higher inpatient costs (adjusted cost ratio 1.46, 95% CI 1.44 to 1.47) compared to those without dysphagia [46].

Description of the intervention and how it might work

Rehabilitation interventions for oropharyngeal dysphagia have four interrelated aims: (1) improve swallowing safety by reducing penetration (when material stays above or contacts the vocal folds) and aspiration (when material enters the trachea and goes below the vocal folds) of food and fluids [47]; (2) improve swallowing efficiency by increasing oral and pharyngeal co-ordination during bolus swallowing; (3) improve overall swallowing ability to reduce the risk of malnutrition; and (4) improve quality of life and active participation.

The first-line treatment for movement disorders in Parkinson's disease is pharmacological: people with the disease take dopaminergic medications (drugs that increase the amount of dopamine in the brain), which are recognised as fundamental therapy for movement disorders. The effects of these medications on swallowing function are controversial. It seems that they do not directly affect swallowing, but they reduce rigidity, festination, or freezing episodes, which impact eating processes [11, 48, 49, 50].

Rehabilitation interventions for oropharyngeal dysphagia are complex interventions in keeping with the definition provided by the Medical Research Council [51, 52]. They can be categorised into two main groups: behavioural treatments and stimulation interventions.

Behavioural treatments

Behavioural swallowing treatments involve compensatory strategies, exercise programmes, or both [5, 48, 53, 54].

Compensatory strategies include postural manoeuvres, changing swallowing behaviour (e.g. taking a smaller bolus), and changes to food and/or fluid consistency. These compensatory strategies can result in an immediate positive effect on swallowing [55, 56, 57, 58]. These strategies are usually applied at the first sign of oropharyngeal dysphagia. Modification of food/fluid consistencies is typically implemented to reduce difficulties in oral-pharyngeal swallow co-ordination. Thickened liquids are thought to reduce the risk of pre-swallowing entry of fluid into the airway. Recent systematic reviews on the effects of bolus viscosity on the safety and efficiency of swallowing concluded that, although increasing bolus viscosity can immediately impact the safety of swallowing, increasing fluid viscosity may also increase residue in the pharynx and heighten the risk for post-swallow aspiration. It can also have long-term effects on people's health (for example, by leading to malnutrition, dehydration, and urinary tract infections) [59, 60, 61, 62]. Modification of food to include softer, easier-to-chew foods may help with swallowing in people with masticatory difficulties [55, 56, 57, 58].

Exercise programmes aim to improve swallowing safety and efficiency by altering individuals' physiology [63, 64]. Such exercise programmes can be broadly divided into two groups.

(1) Training that does not involve the act of swallowing but has positive effects on associated swallowing function. Examples of these exercises are respiratory and voice training [25, 65, 66], or tongue strength exercises [67]. Respiratory and voice training seems to increase swallowing safety [25, 68, 69]. Accordingly, expiratory muscle strength training (EMST) has been applied to increase the airway protection mechanism during swallowing in people with Parkinson's disease [25].

(2) Training that aims directly at increasing the skills and co-ordination involved in swallowing [48, 53, 63]. These exercises incorporate the neuroplasticity principle of 'use it or lose it' [64, 70]. According to this principle, swallowing training may stimulate the cortical and subcortical areas responsible for deglutition (the action or process of swallowing) to promote recovery. These exercises include skill swallowing training or video-assisted therapy [39, 40]. Some of these rehabilitative interventions also include the use of biofeedback, which seems to be specifically beneficial in the recovery of swallowing function in people with Parkinson's disease and oropharyngeal dysphagia [10, 71].

Stimulation interventions

In recent years, there has been an increasing interest in stimulation to improve swallowing. There are different types of stimulation, which can be divided into central and peripheral stimulation. Central stimulation involves devices that stimulate different parts of the brain, such as deep brain stimulation (DBS), transcranial magnetic stimulation (TMS), and transcranial direct current stimulation (tDCS). DBS is a type of central stimulation that continuously delivers electric current to the basal ganglia's deep structures. It is a recognised treatment for reducing motor symptoms in individuals with Parkinson's disease. Nevertheless, it seems not to increase swallow function and swallowing safety [12, 72]. TMS and tDCS are brain stimulation interventions that modulate brain activities through electromagnetic and electrical induction. Some studies have suggested that repetitive TMS (rTMS) can induce long-lasting neural plasticity changes in the swallowing motor cortex, which plays a key role in the recovery of swallowing function [12, 73, 74]. Several other stimulation interventions have emerged in recent years; in particular, pharyngeal electrical stimulation (PES) and neuromuscular electrical stimulation (NMES). PES stimulates the pharyngeal mucosa directly through intraluminal electrodes [12, 74]. NMES stimulates the skin around the neck or facial area through surface electrodes [12, 75]. Within the subgroup of peripheral stimulation interventions, there are peripheral sensory stimulations. These interventions include tactile, thermal, or other multimodal sensory stimulation approaches [47, 76, 77] and carbonation [78, 79, 80]. These deliver stimulation to peripheral nerves or structures that are involved in swallowing in order to initiate the pharyngeal swallow.

Recent systematic reviews have revealed that rehabilitation interventions for oropharyngeal dysphagia may potentially reduce the risk of aspiration and penetration of food and fluids into the trachea, by improving the timing of pharyngeal swallow initiation and hyolaryngeal excursion (closure of the airway through the movement of the hyoid bone and larynx) during swallowing [10, 12, 37, 41, 48]. These reviews also found that rehabilitation interventions improved the quality of life of people with Parkinson's disease. In a recent scoping review, Hirschwald and colleagues identified core outcomes used for assessing dysphagia interventions in Parkinson's disease. The included studies on rehabilitation interventions for oropharyngeal dysphagia in Parkinson's disease primarily assessed the following outcomes [42]:

- swallowing safety: reducing the risk of penetration and aspiration;
- swallowing-related physiology: increasing oral bolus control and transport, reducing the timing of oropharyngeal swallow;
- swallowing efficiency: reducing post-swallow oral pharyngeal residue and pooling, and reducing piecemeal deglutition;
- swallowing-related quality of life.

In addition, some studies reported the benefit of rehabilitation in enhancing airway protection and reducing aspiration pneumonia [25, 81]. Swallowing interventions also increased saliva management in some studies, although few have measured it as an outcome [37, 42, 82].

Some studies showed that swallowing treatment led to positive changes in neurological symptoms and voice [69, 83]. Furthermore, rehabilitation for oropharyngeal dysphagia seemed to impact life functioning, increasing perceived health status, well-being, and quality of life [37, 41]. One study reported death and hospitalisation rates as outcome measures in people with Parkinson's disease [62].

Why it is important to do this review

Oropharyngeal dysphagia is considered one of the main causes of death in people with Parkinson's disease, impacting negatively on their quality of life, activities of daily living, and participation in social life. People with Parkinson's disease, their caregivers, and healthcare professionals require robust evidence about the most effective rehabilitation interventions for oropharyngeal dysphagia.

The last Cochrane systematic review on swallowing rehabilitation in people with Parkinson's disease was published in 2001 [84]. Deane and colleagues concluded that there was no evidence to support or refute the efficacy of non-pharmacological swallowing treatments for oropharyngeal dysphagia in Parkinson's disease, and they included no randomised controlled trials (RCTs) in their review [84]. In the last two decades, RCTs on rehabilitation interventions for oropharyngeal dysphagia in Parkinson's disease have been conducted, exploring a range of different approaches, from behavioural treatment to neuro-stimulation. Given the increasing prevalence of Parkinson's disease and the severe complications of oropharyngeal dysphagia in this population, an up-to-date summary and synthesis of the available evidence on the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in Parkinson's disease was needed.

OBJECTIVES

To assess the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in improving swallowing safety and efficiency in people with Parkinson's disease.

METHODS

We followed guidance in the Methodological Expectations of Cochrane Intervention Reviews (MECIR) handbook when conducting the review [85], and PRISMA 2020 for the reporting [86].

Due to the limited number of studies included in the meta-analyses, we did not generate funnel plots to assess possible publication bias, nor did we conduct subgroup or sensitivity analyses as planned.

Criteria for considering studies for this review

Types of studies

We included eligible randomised controlled trials (RCTs) and cluster-randomised controlled trials (cluster-RCTs). We included eligible cross-over randomised studies only if we could obtain separate data for the first study period from the study's publication or authors. We excluded non-randomised studies of interventions, cross-sectional studies, case-control studies, and observational uncontrolled study designs.

Types of participants

We included adults (aged 18 years and over) with a confirmed diagnosis of idiopathic Parkinson's disease with any duration or severity, according to guidelines defined by the Movement Disorders Society [87]. We included participants experiencing oropharyngeal dysphagia as a consequence of Parkinson's disease. In the case of a mixed population, we contacted the study's authors to request separate data for people with Parkinson's disease, and we excluded these studies if separate data were not available.

We excluded studies involving participants with atypical Parkinsonism and other neurological diseases potentially causing oropharyngeal dysphagia.

Types of interventions

We included any rehabilitation intervention for oropharyngeal dysphagia. These included any of the following, alone or in combination with each other, with or without concurrent pharmacological treatments:

- behavioural interventions, such as swallowing exercise programmes, compensatory manoeuvres, and diet modifications;
- physical stimulation, such as thermal or tactile stimulation;
- neuromuscular electrical stimulation (NMES);
- pharyngeal electrical stimulation (PES);
- transcranial direct current stimulation (tDCS);
- transcranial magnetic stimulation (TMS);
- deep brain stimulation (DBS).

Eligible comparators could include:

- no intervention (e.g. waiting-list control group);
- usual care: this may be described as standard treatment or usual treatment, and refers to the treatment usually available in the study locality. Usual care might include, for example, modified food consistencies or compensatory postures used by participants as a usual feeding method, compensatory manoeuvres, diet modifications, or combinations of these;
- placebo/sham stimulation or attention control intervention;
- any other rehabilitation intervention: participants in the comparator group may receive another swallowing rehabilitation intervention, which is intended to influence the main outcomes of interest, but consists of different therapeutic components to the experimental intervention.

We excluded pharmacological interventions alone.

Outcome measures

We analysed the critical and important outcomes at the end of treatment (according to the individual studies' duration), and at the latest available follow-up. We did not use outcome reporting as a basis for including or excluding studies. For each trial, we analysed only the outcomes listed below and not all the outcomes reported in the trial.

In case of multiple outcome measures for the same outcome, we selected the one that was most clinically meaningful and that had evidence of reliability and validity, if available.

Critical outcomes

- Swallowing efficiency (e.g. the Yale Pharyngeal Residue Severity Rating Scale [88]; Bolus Residue Scale [89]; Eisenhuber scale [90])
- Swallowing safety (e.g. Penetration-Aspiration Scale (PAS) [47])

Important outcomes

- Dysphagia severity using validated outcome measures (e.g. the Dysphagia Outcome and Severity Scale (DOSS) [91]; the

Dysphagia Severity Scale (DSS) [92]; the Functional Oral Intake Scale (FOIS) [93])

- Saliva management (change in frequency and severity of drooling and salivary pooling)
- Quality of life (e.g. Swallowing Quality of Life (SWAL-QoL) [94])
- Respiratory outcomes (including chest infection or aspiration pneumonia; measurements of cough, such as peak expiratory flow rate, cough peak flow, reflex cough threshold, and reflex cough peak [23, 24, 26, 68])
- Nutritional and dietary outcome weight changes, using measures as well as validated scales (e.g. Malnutrition Universal Screening Tool (MUST) [95])
- Adverse events (hospitalisations, mortality)

Search methods for identification of studies

Electronic searches

We searched the following databases on 26 September 2025:

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 9);
- MEDLINE (PubMed) (1946 to 26 September 2025);
- Embase (Embase.com) (1947 to 26 September 2025);
- Cumulative Index to Nursing and Allied Health Literature (CINAHL) EBSCO (1937 to 26 September 2025);
- Web of Science (1900 to 26 September 2025);
- Speech Pathology Database for Best Interventions and Treatment Efficacy (SpeechBite) (speechbite.com) (2008 to 26 September 2025).

A review author (MJDF) developed detailed search strategies for each database (see Supplementary material 1). We imposed no language or publication date restrictions on the search strategies.

Searching other resources

We also searched the US National Institutes of Health Ongoing Trials register, ClinicalTrials.gov (www.clinicaltrials.gov) and the World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (trialsearch.who.int) for ongoing studies and recently completed trials up to 26 September 2025, and OpenGrey (System for Information on Grey Literature in Europe) up to 5 February 2024, as the site has been archived (see Supplementary material 1). We manually reviewed the reference lists of all included studies and of any relevant systematic reviews. We performed a forward citation search of all included studies to identify additional studies not retrieved through electronic searches, as well as any post-publication amendments. Similarly, we searched the Retraction Watch Database for all included studies.

We contacted the investigators of any known ongoing studies to try to obtain any relevant unpublished data.

Data collection and analysis

Selection of studies

Two review authors (IB and MJDF) independently screened titles and abstracts. We resolved any disagreements through discussion with a third review author (MW). We retrieved the full-text articles of studies that appeared to meet the inclusion criteria. Two review authors (IB and MJDF) independently screened the articles for inclusion in the review. In the event of a disagreement between authors, we involved a third review author (MW) to reach a final decision. We used Covidence software for the screening process [96]. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram [86]. We listed studies that were ineligible for inclusion in the 'Characteristics of excluded studies' table, together with reasons for their exclusion.

Data extraction and management

Using Covidence, we recorded study data on data extraction forms that we developed and tested in advance. Two review authors (IB and MJDF) independently extracted study information, and a third author (SGL) checked the extracted information against the original study publications. Two review authors (IB and MJDF) independently extracted outcome data and compared the extractions. We resolved any disagreements through discussion and involved a third review author (MW) if necessary. If contact details were available, we contacted the authors of published trials to clarify trial data or to provide additional information if required.

We extracted the following study characteristics.

- Methods: study design, study duration, number of study centres and location, study setting (i.e. inpatients, outpatients, home-based), withdrawals, and study date.
- Participants: number of participants, mean age, age range, gender, disease duration, the severity of the condition according to the Hoehn and Yahr (H&Y) Scale [97] or Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [98], swallowing-related baseline data, inclusion and exclusion criteria.
- Interventions: experimental intervention, comparison, co-interventions, concomitant pharmacological treatments, duration, frequency, and dose of the interventions.
- Outcomes: primary and secondary outcomes specified and collected, and time points reported.
- Results: the number of events and participants per treatment group for dichotomous outcomes, means and standard deviations, and the number of participants per treatment group for continuous outcomes for each time point. If both final values and change-from-baseline values were reported for the same outcome, we extracted the final values; if both unadjusted and adjusted values for the same outcome were reported, we extracted the adjusted values. If data were analysed based on an intention-to-treat (ITT) sample and another sample (e.g. per-protocol, as-treated), we extracted the data from the per-protocol analysis.
- Characteristics of the trial design, as outlined below in Risk of bias assessment in included studies.
- Notes: funding for the trial, notable declarations of interest of trial authors, and trial registration number.

Risk of bias assessment in included studies

Two review authors (IB and MJDF) independently assessed the risk of bias using the Cochrane risk of bias 2 tool (RoB 2) [99], as described in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [100]. We resolved any disagreements through discussion and by involving a third review author (CA), if necessary. We assessed the outcomes specified for inclusion in the summary of findings tables at the end of treatment and the latest available follow-up. For this review, we were interested in the effect of assignment to the interventions at baseline, regardless of whether the interventions were received as intended (i.e. the 'intention-to-treat' effect).

We considered the following domains:

- bias arising from the randomisation process;
- bias introduced by deviations from intended interventions;
- bias arising from missing outcome data;
- bias in measurement of the outcome;
- bias in selection of the reported result.

We judged each domain as 'low risk of bias', 'high risk of bias', or 'some concerns'. The judgements were facilitated by an algorithm that maps responses to the signalling questions with response options of 'yes', 'probably yes', 'probably no', 'no', and 'no information'. We presented an overall risk of bias judgement for each result based on the assessments across the five domains, considering the worst risk of bias in any of the domains. The overall risk of bias was judged as 'low risk of bias', 'high risk of bias', or 'some concerns'. We performed and documented the risk of bias assessment using the RoB 2 Excel tool (available at www.riskofbias.info).

Measures of treatment effect

For dichotomous outcome data, we calculated effect sizes as risk ratios (RR) with their respective 95% confidence intervals (CIs).

For continuous outcomes, when study authors used the same measures to assess the same outcome, we calculated mean differences (MDs) with 95% CIs. When study authors used different measures to assess the same outcome, we calculated standardised mean differences (SMDs) with 95% CIs if we deemed that pooling the results of different scales that measure the same outcome was appropriate and meaningful. We classified the effect size measured by the SMD as trivial or no effect (if the SMD was between 0.00 and 0.19), small (SMD between 0.20 and 0.49), medium (SMD between 0.50 and 0.79), and large (SMD 0.80 or greater) [101].

Unit of analysis issues

Studies with multiple treatment groups

If a study compared two or more treatment groups eligible for inclusion in the review, we labelled the arms separately in pair-wise analyses. To avoid including data for controls more than once in the same comparison, we divided the number of events and the number of participants in the control group into equal parts. Where suitable, we combined multiple treatment arms representing the same intervention category, as defined for the purposes of the review, to create a single pair-wise comparison, according to the methods recommended in Chapters 6 and 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [102, 103], and we excluded irrelevant treatment arms.

Cross-over trials

We included relevant eligible randomised cross-over trials in the review. Still, we only used data gathered during the first period of the study, up to the point of crossing over. This avoided any problems associated with any carry-over effect from the first to the second study period.

Dealing with missing data

If data on intervention groups, summary data, standard deviations on specific outcomes, or study design were missing, we contacted study authors or data owners to request these data.

For continuous outcomes, we calculated the MD or SMD based on the number of participants analysed at the selected time point. If the study authors did not report the number of participants analysed for each time point, we used the number of randomised participants in each group at baseline.

Where possible, we calculated missing standard deviations from other statistics, such as standard errors, confidence intervals, or P values, according to the methods recommended in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [102].

Reporting bias assessment

We planned to investigate the possibility of reporting biases, including publication bias, by visually assessing funnel plots for asymmetry where more than 10 studies were included in the meta-analysis [104].

In order to minimise the impact of reporting biases, we conducted comprehensive searches of multiple databases and other sources, including clinical trial registries, to identify any unpublished studies (see Search methods for identification of studies). We also looked for outcome reporting bias in studies by recording all trial outcomes, planned and reported, and noting the absence of anticipated outcomes or less detailed reporting of non-significant outcomes. We contacted study authors to try to obtain any missing data.

Synthesis methods

We pooled the results from the individual trials in meta-analyses, where possible (i.e. good comparability of types of participants, intervention, comparator, and outcomes between trials), with the DerSimonian and Laird method [105]. We employed a random-effects model, as we expected a certain degree of clinical heterogeneity amongst the included studies.

When meta-analysis was not appropriate due to unacceptable heterogeneity, we presented a narrative summary of study results, summarising effect estimates when possible, following guidance in Chapter 12 of the *Cochrane Handbook for Systematic Reviews of Interventions* [106].

We planned to pool data if studies were similar in terms of their study populations, interventions, and outcomes (PICO elements). We assessed the presence of clinical heterogeneity within each pair-wise comparison by comparing the trial and study population characteristics across all eligible trials. For pair-wise analyses, we inspected forest plots visually to detect heterogeneity. We used the I^2 test to assess statistical heterogeneity, using the following rough guide to the interpretation of heterogeneity values set out in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [107]:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;
- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: represents considerable heterogeneity.

These overlapping intervals reflect that the interpretation of the I^2 statistic depends on the value size and direction of the treatment effect and variance of the I^2 estimate.

Investigation of heterogeneity and subgroup analysis

To investigate heterogeneity, we planned to perform the following subgroup analyses for the critical outcomes (i.e. swallowing efficiency and swallowing safety).

- Severity of swallowing safety at baseline using instrumental examination (Penetration-Aspiration Scale (PAS) ≤ 5 versus PAS > 5) [47]. We set the cut-off at PAS ≤ 5 versus PAS > 5, because it distinguishes between food penetration events and food aspiration events [42].
- Parkinson's disease onset: early onset at 50 years or younger versus older than 50 years. With advancing age, individuals often develop substantial impairments that may affect the effectiveness of swallowing interventions [108].
- Severity of Parkinson's symptoms/stage of disease according to the Hoehn and Yahr Scale (Mild 1-2; Moderate/Severe 3-5) [97, 109], or the MDS-UPDRS Scale (Part III: Mild/Moderate ≤ 58; Severe ≥ 59) [98, 109]. In advanced stages of Parkinson's disease, pronounced motor and cognitive impairments can have a substantial impact on the selection and effectiveness of swallowing interventions.

We planned to compare subgroups using the formal test for subgroup differences in Review Manager [110].

Equity-related assessment

We did not investigate equity-related characteristics in this review.

Sensitivity analysis

We planned to check the robustness of the results of our primary analyses for the critical outcomes (i.e. swallowing efficiency and swallowing safety) by conducting sensitivity analyses restricted to the results of studies we rated as having either 'low risk' or 'some concerns' of bias overall.

Certainty of the evidence assessment

We produced summary of findings tables to present summary effect estimates for the following outcomes:

- swallowing efficiency;
- swallowing safety;
- dysphagia severity;
- saliva management;
- quality of life;
- respiratory outcomes;
- adverse events.

We presented summary effect estimates at the end of treatment (according to the individual studies' duration) and at the latest available follow-up.

Considering the multicomponent characteristics of rehabilitation [111, 112], we could not a priori specify the most important comparisons. Therefore, we presented those we considered to be clinically relevant in the summary of findings tables.

We followed the GRADE approach in preparing the tables. Two review authors (CA and SGL) independently performed GRADE assessments, and resolved any disagreements through discussion. We assessed the evidence against the following domains:

- study limitations (risk of bias);
- imprecision of results (wide confidence intervals for treatment effect);
- heterogeneity (large I^2 value);
- indirectness of evidence (variations in participants, interventions, comparisons, and outcomes);
- publication bias.

According to the GRADE approach, we assessed the overall certainty of the evidence for each outcome as 'high', 'moderate', 'low', or 'very low'.

- High certainty: we are confident that the true effect lies close to that of the estimate of the effect.
- Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimated effect.

We followed the guidelines in the *Cochrane Handbook for Systematic Reviews of Intervention*s, Chapters 14 and 15 [113, 114], for interpreting results, distinguishing a lack of evidence of effect from a lack of effect. We avoided making recommendations for practice.

Consumer involvement

Consumers were not involved in this review due to limited resources.

RESULTS

Description of studies

We provide a full description of the included studies in Supplementary material 2 and a summary of trial and participant characteristics in Table 1.

Results of the search

Our literature search yielded 2299 potentially relevant references (119 PubMed, 212 CENTRAL, 122 CINAHL, 934 Embase, 78 SpeechBite, 571 Web of Science, 34 OpenGrey, 135 ClinicalTrials.gov, 94 WHO ICTRP). After removing duplicates, two authors (IB, MJDF) screened 1655 for titles and abstracts. We excluded 1543 records, leaving a total of 112 records for full-text analysis. After full-text review, 43 studies (49 records) were excluded (Figure 1). We included 18 studies (43 records) in the review (Baijens 2013 [115, 116]; Cebi 2025 [117, 118, 119, 120]; Claus 2021 [121, 122, 123]; Dashtelei 2024 [124, 125]; Feng 2019 [126]; Heijnen 2012 [127]; Khan 2025 [128, 129]; Khedr 2019 [130, 131]; Logemann 2008 [132, 133, 134]; Manor 2013 [135, 136]; Mohseni 2023 [137]; Park 2018 [138, 139]; Plaza 2022 [140]; Riboldazzi

2020 [141, 142]; Troche 2010 [143, 144, 145, 146, 147, 148, 149, 150]; Troche 2023 [151, 152, 153, 154]; Xie 2018 [155, 156]; Zhang 2025 [157]) (Supplementary material 2). We listed five studies (five records) as awaiting classification (Supplementary material 4) and identified 13 ongoing studies (15 records) (Supplementary material 5).

Figure 1. PRISMA study flow diagram

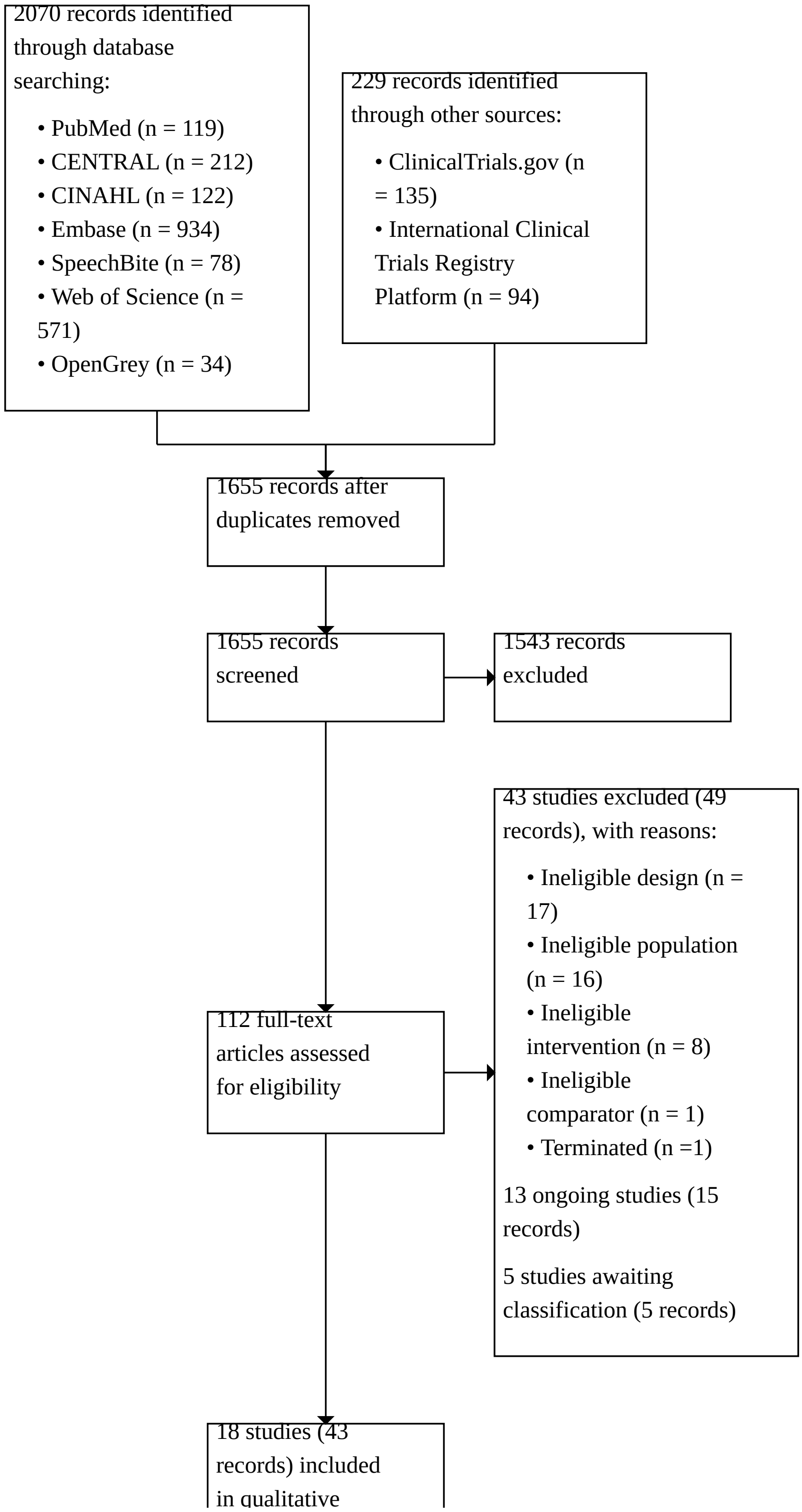
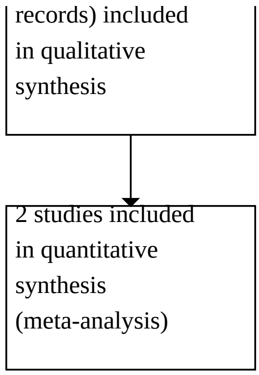


Figure 1. (Continued)



Included studies

We included 17 parallel-group RCTs (Baijens 2013; Cebi 2025; Claus 2021; Dashtelei 2024; Feng 2019; Heijnen 2012; Khan 2025; Khedr 2019; Logemann 2008; Manor 2013; Mohseni 2023; Park 2018; Plaza 2022; Riboldazzi 2020; Troche 2010; Troche 2023; Zhang 2025) and one cross-over randomised study (Xie 2018), with a total of 1216 participants. Five of the 18 included studies had three eligible intervention arms (Baijens 2013; Heijnen 2012; Logemann 2008; Mohseni 2023; Xie 2018). We found no cluster-randomised trials. The characteristics of the included studies are described in Supplementary material 2.

We contacted the authors of 14 included studies for additional information. We received a response including additional trial information or outcome data from the authors of eight studies (Baijens 2013; Cebi 2025; Claus 2021; Khan 2025; Logemann 2008; Riboldazzi 2020; Troche 2010; Troche 2023). The authors of two studies said that access to the database was not possible as the permission to retrieve the data was no longer available (Baijens 2013; Logemann 2008). Authors of six studies did not reply (Feng 2019; Heijnen 2012; Khedr 2019; Manor 2013; Xie 2018; Zhang 2025).

Country

Four studies were conducted in the USA (Logemann 2008; Troche 2010; Troche 2023; Xie 2018); two each in China (Feng 2019; Zhang 2025), Germany (Cebi 2025; Claus 2021), the Netherlands (Baijens 2013; Heijnen 2012) and Iran (Dashtelei 2024; Mohseni 2023); and one each in Italy (Riboldazzi 2020), Chile (Plaza 2022), Pakistan (Khan 2025), the Republic of Korea (Park 2018), Egypt (Khedr 2019), and Israel (Manor 2013).

Population

All studies were published between 2008 (Logemann 2008) and 2025 (Cebi 2025; Khan 2025; Zhang 2025). Sample sizes of the included studies ranged from 11 (Xie 2018) to 360 participants (Logemann 2008). A total sample size of 1216 participants with Parkinson's disease and dysphagia were included in the RCTs. Most participants were male (71.58%). Two studies did not report participants' gender (Khedr 2019; Logemann 2008). The mean age of participants ranged from 58.8 to 76.2 years. The mean time since diagnosis ranged from 5.7 years to 10.1 years, although this information was not reported in three studies (Heijnen 2012; Khan 2025; Troche 2010). Fifteen studies reported disease severity using the Hoehn and Yahr Scale, which ranged from 2 to 4; only three studies did not include this information (Cebi 2025; Feng 2019; Khan 2025). Four studies also administered the MDS-UPDRS (Cebi 2025; Claus 2021; Mohseni 2023; Troche 2023).

Interventions

The included studies looked at various forms of swallowing rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson's disease. Intervention details are provided in Supplementary material 2. Ten studies explored behavioural interventions, while eight studies tested one or more stimulation interventions.

Behavioural interventions

Ten studies tested behavioural interventions in 770 participants (Claus 2021; Feng 2019; Khan 2025; Logemann 2008; Manor 2013; Mohseni 2023; Plaza 2022; Riboldazzi 2020; Troche 2010; Troche 2023). Four studies investigated expiratory training (Claus 2021; Riboldazzi 2020; Troche 2010; Troche 2023). Two studies investigated compensatory manoeuvres and diet modifications (Logemann 2008; Manor 2013). Two studies involved voice training (Khan 2025; Mohseni 2023). Two studies comprised tongue strength training using biofeedback (Feng 2019; Plaza 2022).

Stimulation interventions

Two studies tested repetitive transcranial magnetic stimulation (rTMS) in 90 participants (Khedr 2019; Zhang 2025). One study assessed transcranial direct current stimulation (tDCS) in 29 participants (Dashtelei 2024). Two studies tested deep brain stimulation (DBS) in 351 participants (Cebi 2025; Xie 2018). Four studies tested neuromuscular electrical stimulation (NMES) in 292 participants (Baijens 2013; Heijnen 2012; Park 2018; Zhang 2025).

Comparisons

Seven studies compared the swallowing experimental intervention with a sham intervention (Cebi 2025; Claus 2021; Dashtelei 2024; Khedr 2019; Park 2018; Troche 2010; Xie 2018).

Ten studies compared the experimental intervention to a behavioural swallowing treatment (Baijens 2013; Feng 2019; Heijnen 2012; Khan 2025; Logemann 2008; Manor 2013; Mohseni 2023; Plaza 2022; Riboldazzi 2020; Troche 2023); of these, three included a behavioural swallowing treatment comprising compensatory manoeuvres, diet modifications, or combinations of these (Logemann 2008; Manor 2013; Riboldazzi 2020). One study compared the experimental intervention to a respiratory treatment (Troche 2023), while another compared speech therapy to music therapy (Mohseni 2023). One study compared rTMS to NMES (Zhang 2025).

Five were multi-arm studies with three groups: two studies incorporated three different swallowing behavioural interventions (Logemann 2008; Mohseni 2023), two compared two types of neuromuscular electrical stimulation swallowing treatments against traditional swallowing treatment (Baijens 2013; Heijnen 2012), and one study investigated two different mode of DBS stimulation (Xie 2018).

Outcomes

Outcome assessment was conducted after treatment in all included studies except Xie 2018. Six studies included a follow-up assessment after one (Dashtelei 2024), three (Claus 2021; Logemann 2008), six (Manor 2013), seven (Mohseni 2023), and 14.5 months (Xie 2018), respectively.

Swallowing efficiency

Seven studies assessed swallowing efficiency (Baijens 2013; Claus 2021; Khedr 2019; Manor 2013; Park 2018; Plaza 2022; Troche 2010), and four studies assessed post-swallowing residue (Baijens 2013; Claus 2021; Khedr 2019; Manor 2013). The timing of oropharyngeal swallow components was evaluated in three studies (Baijens 2013; Khedr 2019; Troche 2010). The swallowing-related hyoid bone movements were analysed in two studies (Park 2018; Troche 2010). Swallowing efficiency was evaluated at different time points across studies: Baijens 2013 at 15 days (end of treatment); Claus 2021 at four weeks (end of treatment) and at 3-month follow-up; Khedr 2019 at two weeks (end of treatment); Manor 2013 at four months (end of treatment) and 6-month follow-up; Park 2018, Plaza 2022, and Troche 2010 at four months (end of treatment).

Swallowing safety

Nine studies assessed swallowing safety (Baijens 2013; Cebi 2025; Claus 2021; Dashtelei 2024; Khedr 2019; Logemann 2008; Manor 2013; Park 2018; Xie 2018). Of these, seven used the Penetration-Aspiration Scale (PAS) [47] (Baijens 2013; Cebi 2025; Claus 2021; Dashtelei 2024; Khedr 2019; Logemann 2008; Park 2018). Xie 2018 assessed the frequency of aspirations, and Manor 2013 adopted a non-validated clinical scale. Swallowing safety was evaluated at different time points across studies: Baijens 2013 at 15 days (end of treatment); Cebi 2025 at eight months (end of treatment); Claus 2021 at four weeks (end of treatment) and at 3-month follow-up; Dashtelei 2024 at two weeks (end of treatment) and 1-month follow-up; Khedr 2019 at two weeks (end of treatment); Logemann 2008 at the end of a single session of treatment; Manor 2013 at four months (end of treatment) and 6-month follow-up; Park 2018 at four months (end of treatment); and Xie 2018 at 14.5-month follow-up.

Dysphagia severity

Twelve studies investigated dysphagia severity (Cebi 2025; Claus 2021; Dashtelei 2024; Feng 2019; Heijnen 2012; Khan 2025; Khedr 2019; Manor 2013; Mohseni 2023; Plaza 2022; Xie 2018; Zhang 2025). The Swallowing Disturbance Questionnaire (SDQ) was administered in six RCTs [158] (Claus 2021; Dashtelei 2024; Khedr 2019; Manor 2013; Mohseni 2023; Xie 2018), the Dysphagia Handicap Index (DHI) in three RCTs [159] (Dashtelei 2024; Khedr 2019; Mohseni 2023), the Functional Oral Intake Scale (FOIS) in five RCTs [93] (Cebi 2025; Heijnen 2012; Khan 2025, Plaza 2022; Zhang 2025), and the Dysphagia Severity Scale (DSS) [92] and Standardised Swallowing Assessment (SSA) in one RCT each (Heijnen 2012; Feng 2019), respectively. Dysphagia severity was evaluated at different time points across studies: Cebi 2025 at eight weeks (end of treatment); Claus 2021 at four weeks (end of treatment) and at 3-month follow-up; Dashtelei 2024 at two weeks (end of treatment) and 1-month follow-up; Feng 2019 at four weeks (end of treatment); Heijnen 2012 at 34 days (end of treatment); Khan 2025 at 12 weeks (end of treatment); Khedr 2019 at two weeks (end of treatment); Manor 2013 at four weeks (end of treatment) and at 6-month follow-up; Mohseni 2023 at four weeks (end of treatment) and 3-month follow-up; Plaza 2022 at four weeks (end of treatment); Xie 2018 at 14.5-month follow-up; and Zhang 2025 at 20 days (end of treatment).

Saliva management

Two studies assessed salivary management (Feng 2019; Zhang 2025). Feng 2019 applied the Drooling Severity and Frequency Scale (DSFS; both subscales) at four weeks (end of treatment); Zhang 2025 used the Repetitive Saliva Swallowing Test (RSST) at 20 days (end of treatment).

Quality of life

Six studies assessed quality of life (Cebi 2025; Claus 2021; Heijnen 2012; Manor 2013; Riboldazzi 2020; Troche 2010). Of these, five studies used the Swallowing Quality of Life (SWAL-QoL) instrument [94] (Cebi 2025; Claus 2021; Heijnen 2012; Manor 2013; Troche 2010), and one study adopted the Parkinson's Disease Questionnaire (PDQ-39) [160] (Riboldazzi 2020). Quality of life was evaluated at different time points across studies: Cebi 2025 at eight weeks (end of treatment); Claus 2021 at four weeks (end of treatment) and at 3-month follow-up; Heijnen 2012 at 34 days (end of treatment); Manor 2013 at four months (end of treatment) and 6-month follow-up; Riboldazzi 2020 at 12-month (end of treatment); and Troche 2010 at four months (end of treatment).

Respiratory outcomes

Two RCTs included respiratory outcome measures (Riboldazzi 2020; Troche 2023), both assessing the Voluntary Cough Peak Expiratory Flow Rate (PEFR). Riboldazzi 2020 also evaluated the Forced Vital Capacity (FVC), Forced Expiratory Volume on the 1st second (FEV 1), Peak Expiratory Flow (PEF), while Troche 2023 assessed Maximum Expiratory Pressure (MEP), the Cough Expired Volume (CEV), and Reflex Cough Peak Expiratory Flow Rate. Respiratory outcomes were evaluated at different time points by the two studies: Riboldazzi 2020 at 12 months (end of treatment) and Troche 2023 at five weeks (end of treatment).

Nutritional and dietary outcomes

None of the included studies evaluated weight changes or dietary information.

Adverse events

Two studies assessed adverse events (Logemann 2008; Riboldazzi 2020). They investigated mortality and pneumonia rates, evaluated body temperature (as an indirect marker of possible infections), and measured the hospitalisation rate (including access to the emergency department, general practitioner, or both).

Synthesis methods

Of the 18 included studies, twelve were suitable for inclusion in quantitative synthesis. We were able to pool data from two studies for only one comparison – expiratory muscle strength training versus sham treatment – and one outcome (swallowing safety at the end of treatment) (Claus 2021; Troche 2023). Because of clinical heterogeneity across the interventions and comparators, we calculated individual effect estimates for the remaining comparisons and outcomes (Claus 2021; Dashtelei 2024; Feng 2019; Khedr 2019; Manor 2013; Mohseni 2023; Park 2018; Plaza 2022; Riboldazzi 2020; Troche 2010; Troche 2023; Zhang 2025).

The data from six studies were not available in a format suitable for pooling (Baijens 2013; Cebi 2025; Heijnen 2012; Khan 2025; Logemann 2008; Xie 2018). Instead, we summarised these data in Table 2.

Excluded studies

We excluded 43 studies (49 records) at the full-text screening stage (see Figure 1 and Supplementary material 3) for the following reasons:

- ineligible study design (17 studies);
- ineligible participants (16 studies);
- ineligible intervention (eight studies);
- ineligible comparator (one study);
- terminated study (withdrawn for ethical issues) (one study).

Ongoing studies

We identified 13 ongoing studies (15 records) (Supplementary material 5). They were all trials registered in international trial registries: six in ClinicalTrials.gov (NCT05700825 [161]; NCT05837520 [162]; NCT06328296 [163]; NCT06370195 [164]; NCT06791590 [165]; NCT07153692 [166]), two in the Chinese Clinical Trial Registry (ChiCTR) (ChiCTR2200060790 [167]; ChiCTR2300076032 [168]), two in the Brazilian Clinical Trial Registry (Registro Brasileiro de Ensaios Clínicos (ReBEC)) (RBR-3pmvvnx [169]; RBR-5ddt474 [170]), one in the International Standard Randomised Controlled Trial Number (ISRCTN) registry (Dismore 2023 [171, 172]), and two in the University Hospital Medical Information Network Clinical Trials Registry (JPRN-UMIN000056111 [173]; Nakamori 2025 [174, 175]). We contacted the trial investigators of nine studies to verify the study phase, and to enquire whether any results were available (ChiCTR2200060790; ChiCTR2300076032; Dismore 2023; Nakamori 2025; NCT05700825; NCT05837520; NCT06370195; RBR-3pmvvnx; RBR-5ddt474). Three authors confirmed that no unpublished final results were available (Dismore 2023; NCT05700825; NCT05837520). The remaining investigators did not reply. We attempted to contact investigators of one additional trial, but we could not find current contact details (NCT06328296). Three studies stated in the trial registry that they were in the ongoing phase (JPRN-UMIN000056111; NCT06791590; NCT07153692). According to the preliminary information available in the registries, four studies are investigating neurostimulation/neuromodulation interventions (ChiCTR2300076032; Nakamori 2025; NCT06370195; NCT07153692), and nine are focused on behavioural therapies (ChiCTR2200060790; Dismore 2023; JPRN-UMIN000056111; NCT05700825; NCT05837520; NCT06328296; NCT06791590; RBR-3pmvvnx; RBR-5ddt474).

Studies awaiting classification

We identified five studies (five reports) awaiting classification (Supplementary material 4). Of these, three were abstracts presented at international scientific conferences (Akbarkhodjaeva 2025 [176]; Nozaki 2011 [177]; Nozaki 2013 [178]), and two were trial registrations reported as completed studies (ACTRN12618000090213 [179]; NCT01420796 [180]). We attempted to contact the corresponding authors, but none responded. The methodological details were insufficiently reported in the abstracts, preventing definitive classification of these studies.

Risk of bias in included studies

For the studies included in the meta-analysis or effect estimation, we performed risk of bias assessments using the RoB 2 tool for all outcomes at the end of treatment and the latest follow-up. The risk of bias assessments are visually presented within the forest plots with traffic lights. RoB 2 assessment data are available in Supplementary material 6.

Expiratory muscle strength training (EMST) versus sham EMST

Only two studies had data on swallowing safety at the end of treatment (Claus 2021; Troche 2010). We judged Claus 2021 to have an overall high risk of bias related to concerns about potential deviations from intended interventions. We judged Troche 2010 to have an overall low risk of bias. Only single studies provided data on the remaining outcomes. We judged Claus 2021 to have an overall high risk of bias for swallowing safety at the latest follow-up, swallowing efficiency, dysphagia severity, and quality of life at both time points. We judged Troche 2023 to have an overall low risk of bias for peak expiratory flow at the end of treatment.

Video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) versus SLT alone

Only one study provided data for all outcomes. We judged Manor 2013 to have an overall high risk of bias related to the randomisation process.

Neuromuscular electrical stimulation (NMES) plus SLT versus sham NMES plus SLT

Only one study provided data for all outcomes. We judged Park 2018 to have an overall high risk of bias related to the measurement of the outcome.

Tongue strength training plus SLT versus SLT alone

Only one study provided data for dysphagia severity. We judged Plaza 2022 to have an overall high risk of bias related to the measurement of the outcome.

Music therapy plus SLT versus music therapy alone versus SLT alone

Only one study provided data for dysphagia severity. We judged Mohseni 2023 to have some concerns related to the selection of the reported results.

Transcranial direct current stimulation (tDCS) versus sham tDCS

Only one study provided data for all outcomes. We judged Dashtelei 2024 to have an overall high risk of bias related to deviations from intended interventions.

Repetitive transcranial magnetic stimulation (rTMS) vs sham rTMS

Only one study provided data for all outcomes. We judged Khedr 2019 to have an overall high risk of bias related to missing outcome data and the measurement of the outcome.

Expiratory flow acceleration plus SLT versus SLT alone

Only one study provided data for all outcomes. We judged Riboldazzi 2020 to have an overall high risk of bias related to some concerns about missing outcome data and the measurement of the outcome.

Vocalisation training plus SLT versus SLT alone

Only one study provided data for all outcomes. We judged Feng 2019 to have an overall high risk of bias related to the randomisation process and selection of the reported results.

rTMS plus motor imagery therapy versus NMES

Only one study provided data for all outcomes. We judged Zhang 2025 to have an overall high risk of bias related to the randomisation process and selection of the reported results.

Detailed risk of bias assessment data with consensus responses to the signalling questions in the RoB 2 Excel tool are available on reasonable request.

Synthesis of results

We assessed the certainty of evidence and produced summary of findings tables for four clinically relevant comparisons:

- expiratory muscle strength training (EMST) compared to sham EMST (Summary of findings 1);
- video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) compared to SLT alone (Summary of findings 2);
- neuromuscular electrical stimulation (NMES) plus SLT compared to sham NMES plus SLT (Summary of findings 3);
- tongue strength training plus SLT compared to SLT alone (Summary of findings 4).

These comparisons investigated the following outcomes:

- swallowing efficiency;
- swallowing safety;
- dysphagia severity;
- quality of life;
- respiratory outcomes.

We identified seven additional comparisons which investigated the above five outcomes as well as saliva management and adverse events. None of the included studies evaluated weight changes or dietary information.

We did not undertake subgroup analyses because of the small number of studies included in all the analysed comparisons (always fewer than 10) [107].

Comparison 1. Expiratory muscle strength training (EMST) compared to sham EMST

Swallowing efficiency

The evidence is very uncertain about the effects of expiratory muscle strength training on swallowing efficiency after four weeks (MD -4.40, 95% CI -7.50 to -1.30; 1 study, 45 participants; very low-certainty evidence; Analysis 1.1) and at the latest follow-up (after three months) (MD -1.70, 95% CI -4.58 to 1.18; 1 study, 45 participants; very low-certainty evidence; Analysis 1.2).

Swallowing safety

Two studies (113 participants) used the Penetration-Aspiration Scale (PAS) to measure swallowing safety at the end of treatment (four weeks) and were pooled for meta-analysis (Claus 2021; Troche 2010). Although both studies reported using the PAS, we identified differences in scoring formats. Therefore, we applied the standardised mean difference (SMD) in our meta-analysis.

Expiratory muscle strength training may improve swallowing safety after four weeks (SMD -0.66, 95% CI -1.04 to -0.28; I^2 = 0%; 2 studies; 113 participants; very low-certainty evidence; Analysis 1.3), but the evidence is very uncertain. The evidence is also very uncertain about the effects of expiratory muscle strength training on swallowing safety at the latest follow-up (after three months) (MD -0.18, 95% CI -0.69 to 0.33; 1 study, 45 participants; very low-certainty evidence; Analysis 1.4).

Dysphagia severity

The evidence is very uncertain about the effects of expiratory muscle strength training on dysphagia severity after four weeks (MD -2.78, 95% CI -18.04 to 12.48; 1 study, 45 participants; very low-certainty evidence; Analysis 1.5) and at the latest follow-up (after three months) (MD -2.90, 95% CI -16.94 to 11.14; one study, 45 participants; very low-certainty evidence; Analysis 1.6).

Quality of life

The evidence is very uncertain about the effects of expiratory muscle strength training on quality of life after four weeks (MD 14.69, 95% CI -63.52 to 92.90; 1 study, 45 participants; very low-certainty evidence; Analysis 1.7) and at the latest follow-up (after three months) (MD 11.40, 95% CI -67.95 to 90.75; 1 study, 45 participants; very low-certainty evidence; Analysis 1.8).

Peak expiratory flow

Expiratory muscle strength training may result in little to no difference in peak expiratory flow after five weeks (MD -0.50, 95% CI -1.05 to 0.05; 1 study, 58 participants; low-certainty evidence; Analysis 1.9).

Comparison 2. Video-assisted swallowing therapy (VAST) plus speech-language therapy (SLT) versus SLT alone

Dysphagia severity

The evidence is very uncertain about the effects of video-assisted swallowing therapy on dysphagia severity after four weeks (MD -0.70, 95% CI -5.13 to 3.73; 1 study, 42 participants; very low-certainty evidence; Analysis 2.1) and at the latest follow-up (after six months) (MD -4.03, 95% CI -7.85 to -0.21; 1 study, 42 participants; very low-certainty evidence; Analysis 2.2).

Quality of life

Burden impact

The evidence is very uncertain about the effects of video-assisted swallowing therapy on the burden impact subscale of the SWAL-QOL after four weeks (MD 1.62, 95% CI 0.33 to 2.91; 1 study, 42 participants; very low-certainty evidence; Analysis 2.3) and at the latest follow-up (after six months) (MD 1.86, 95% CI 0.82 to 2.90; 1 study, 42 participants; very low-certainty evidence; Analysis 2.4).

Eating desire

Video-assisted swallowing therapy may result in little to no difference in the eating desire subscale of the SWAL-QOL after four weeks (MD 2.53, 95% CI -0.03 to 5.09; 1 study, 42 participants; Analysis 2.5), while it may improve eating desire at the latest follow-up (after six months) (MD 3.69, 95% CI 1.21 to 6.17; 1 study, 42 participants; Analysis 2.6), but this result is informed by one study only.

Fear

Video-assisted swallowing therapy may reduce fear, as measured with the fear subscale of the SWAL-QOL after four weeks (MD 2.40, 95% CI 0.33 to 4.47; 1 study, 42 participants; Analysis 2.7) and at the latest follow-up (after six months) (MD 2.97, 95% CI 0.57 to 5.37; 1 study, 42 participants; Analysis 2.8), but this result is informed by one study only.

Sleep

Video-assisted swallowing therapy may improve sleep, as measured with the sleep subscale of the SWAL-QOL after four weeks (MD 2.74, 95% CI 0.19 to 5.29; 1 study, 42 participants; Analysis 2.9), while it may result in little to no difference in sleep at the latest follow-up (after six months) (MD -0.84, 95% CI -6.82 to 5.14; 1 study, 42 participants; Analysis 2.10), but this result is informed by one study only.

Communication

Video-assisted swallowing therapy may result in little to no difference in the communication subscale of the SWAL-QOL after four weeks (MD 0.47, 95% CI -0.83 to 1.77; 1 study, 42 participants; Analysis 2.11), while it may improve communication at the latest follow-up (after six months) (MD 1.51, 95% CI 0.35 to 2.67; 1 study, 42 participants; Analysis 2.12), but this result is informed by one study only.

Mental health

Video-assisted swallowing therapy may result in little to no difference in the mental health subscale of the SWAL-QOL after four weeks (MD 3.01, 95% CI -0.25 to 6.27; 1 study, 42 participants; Analysis 2.13), while it may improve mental health at the latest follow-up (after six months) (MD 6.31, 95% CI 3.05 to 9.57; 1 study, 42 participants; Analysis 2.14), but this result is informed by one study only.

Social impact

Video-assisted swallowing therapy may result in little to no difference in the social impact subscale of the SWAL-QOL after four weeks (MD 1.23, 95% CI -1.90 to 4.36; 1 study, 42 participants; Analysis 2.15), while it may reduce social impact at the latest follow-up (after six months) (MD 3.12, 95% CI 0.03 to 6.21; 1 study, 42 participants; Analysis 2.16), but this result is informed by one study only.

Symptom frequency

Video-assisted swallowing therapy may reduce symptom frequency, as measured with the symptom frequency subscale of the SWAL-QOL after four weeks (MD 9.90, 95% CI 3.33 to 16.47; 1 study, 42 participants; Analysis 2.17) and at the latest follow-up (after six months) (MD 9.20, 95% CI 3.22 to 15.18; 1 study, 42 participants; Analysis 2.18), but this result is informed by one study only.

Comparison 3. Neuromuscular electrical stimulation (NMES) plus SLT versus sham NMES plus SLT

Swallowing efficiency

The evidence is very uncertain about the effects of neuromuscular electrical stimulation plus SLT on swallowing efficiency after four weeks (MD -2.34, 95% CI -7.16 to 2.48; 1 study, 18 participants; very low-certainty evidence; Analysis 3.1).

Swallowing safety

The evidence is very uncertain about the effects of neuromuscular electrical stimulation plus SLT on swallowing safety after four weeks (MD -1.34, 95% CI -2.32 to -0.36; 1 study, 18 participants; very low-certainty evidence; Analysis 3.2).

Comparison 4. Tongue strength training plus SLT versus SLT alone

Dysphagia severity

The evidence is very uncertain about the effects of tongue strength training plus SLT on dysphagia severity after eight weeks (MD 0.13, 95% CI -0.17 to 0.43; 1 study, 60 participants; very low-certainty evidence; Analysis 4.1).

Comparison 5. Music therapy plus SLT versus SLT alone

Dysphagia severity

Music therapy plus SLT may reduce dysphagia severity after four weeks (MD -6.90, 95% CI -8.85 to -4.95; 1 study, 22 participants; Analysis 5.1) and at the latest follow-up (after three months) (MD -7.54, 95% CI -9.72 to -5.36; 1 study, 22 participants; Analysis 5.2), but this result is informed by one study only.

Comparison 6. Music therapy versus SLT

Dysphagia severity

Music therapy may result in little to no difference in dysphagia severity after four weeks (MD 2.00, 95% CI -1.61 to 5.61; 1 study, 22 participants; Analysis 6.1) and at the latest follow-up (after six months) (MD 1.46, 95% CI -2.36 to 5.28; 1 study, 22 participants; Analysis 6.2), but this result is informed by one study only.

Comparison 7. Transcranial direct current stimulation (tDCS) versus sham tDCS

Swallowing safety

Transcranial direct current stimulation may result in little to no difference in swallowing safety after two weeks (MD 0.01, 95% CI -0.16 to 0.18; 1 study, 20 participants; Analysis 7.1) and at the latest follow-up (after one month) (MD -0.13, 95% CI -0.32 to 0.06; 1 study, 20 participants; Analysis 7.2), but this result is informed by one study only.

Dysphagia severity

Transcranial direct current stimulation may reduce dysphagia severity after two weeks (MD -0.14, 95% CI -0.24 to -0.04; 1 study, 20 participants; Analysis 7.3) and at the latest follow-up (after one month) (MD -0.18, 95% CI -0.32 to -0.04; 1 study, 20 participants; Analysis 7.4), but this result is informed by one study only.

Comparison 8. Repetitive transcranial magnetic stimulation (rTMS) versus sham rTMS

Swallowing efficiency

Repetitive transcranial magnetic stimulation may result in little to no difference in swallowing efficiency after two weeks when testing semi-solid consistency (MD -0.50, 95% CI -1.06 to 0.06; 1 study, 15 participants; Analysis 8.1), fluid consistency (MD -0.50, 95% CI -1.22 to 0.22; 1 study, 15 participants; Analysis 8.2), and solid consistency (MD -0.40, 95% CI -0.96 to 0.16; 1 study, 15 participants; Analysis 8.3), but this result is informed by one study only.

Swallowing safety

Repetitive transcranial magnetic stimulation may result in little to no difference in swallowing safety after two weeks when testing semi-solid consistency (MD 0.00, 95% CI -0.61 to 0.61; 1 study, 15 participants; Analysis 8.4) and fluid consistency (MD 0.10, 95% CI -0.51 to 0.71; 1 study, 15 participants; Analysis 8.5), but this result is informed by one study only. Concerning the solid consistency, the effect size is not estimable due to a reported standard deviation of zero in the sham rTMS group (not estimable; Analysis 8.6).

Comparison 9. Expiratory flow acceleration (EFA) plus SLT versus SLT alone

Quality of life

Expiratory flow acceleration plus SLT may improve quality of life after 12 months (MD -19.00, 95% CI -24.42 to -13.58; 1 study, 20 participants; Analysis 9.1), but this result is informed by one study only.

Peak expiratory flow

Expiratory flow acceleration plus SLT may increase peak expiratory flow after 12 months (MD 3.16, 95% CI 0.73 to 5.59; 1 study, 20 participants; Analysis 9.2), but this result is informed by one study only.

Adverse events – hospitalisation

Expiratory flow acceleration plus SLT may result in little to no difference in hospitalisation after 12 months (OR 0.17, 95% CI 0.01 to 1.88; 1 study, 5 events; Analysis 9.3), but this result is informed by one study only.

Comparison 10. Vocalisation training plus SLT versus SLT alone

Dysphagia severity

Vocalisation training plus SLT may reduce dysphagia severity after four weeks (MD -1.10, 95% CI -1.49 to -0.71; 1 study, 60 participants; Analysis 10.1), but this result is informed by one study only.

Saliva management

Vocalisation training plus SLT may reduce drooling severity (MD -0.87, 95% CI -1.22 to -0.52; 1 study, 60 participants; Analysis 10.2) and drooling frequency after four weeks (MD -0.37, 95% CI -0.62 to -0.12; 1 study, 60 participants; Analysis 10.3), but this result is informed by one study only.

Comparison 11. rTMS plus motor imagery therapy versus NMES

Dysphagia severity

Repetitive transcranial magnetic stimulation plus motor imagery therapy may reduce dysphagia severity after 20 days (MD 1.29, 95% CI 1.00 to 1.58; 1 study, 113 participants; Analysis 11.1), but this result is informed by one study only.

Saliva management

Repetitive transcranial magnetic stimulation plus motor imagery therapy may improve saliva management after 20 days (MD -1.42, 95% CI -1.58 to -1.26; 1 study, 113 participants; Analysis 11.2), but this result is informed by one study only.

Equity assessment

We did not investigate equity-related characteristics in this review.

Reporting biases

Funnel plots were not produced and visually inspected due to lack of data (i.e. fewer than 10 studies included in the meta-analyses).

DISCUSSION

Summary of main results

Key findings

This review included 18 studies (1216 participants) investigating the effects of rehabilitation interventions for oropharyngeal dysphagia in improving swallowing safety and efficiency in people with Parkinson's disease. Data from two studies (113 participants) were included in the meta-analysis. Despite the increase in evidence compared to the previous Cochrane review by Deane and colleagues [84], which found no RCTs, the evidence on the effectiveness of non-pharmacological swallowing interventions for oropharyngeal dysphagia in people with Parkinson's disease remains uncertain.

Are swallowing behavioural treatments more effective than no treatment, sham, or usual treatment?

A total of 10 studies compared behavioural treatment swallowing interventions with sham or usual treatments (Claus 2021; Feng 2019; Khan 2025; Logemann 2008; Manor 2013; Mohseni 2023; Plaza 2022; Riboldazzi 2020; Troche 2010; Troche 2023). The behavioural treatments investigated were expiratory muscle strength training (ESMT), video-assisted swallowing therapy, and tongue strength training using the Iowa Oral Performance Instrument. The evidence suggests the following:

- ESMT, compared with sham ESMT, may improve swallowing safety after four weeks of treatment, but the evidence is very uncertain. Similarly, the evidence is very uncertain about the effects on swallowing safety at three months of follow-up and about the effects on swallowing efficiency, dysphagia severity, and quality of life at all time points. ESMT may result in little to no difference in peak expiratory flow compared with sham ESMT.
- The evidence is very uncertain about the effect of video-assisted swallowing therapy plus speech-language therapy (SLT) versus SLT alone on quality of life and dysphagia severity at all time points.
- The evidence is also very uncertain about the effect of tongue strength training plus SLT on dysphagia severity at the end of eight weeks of treatment when compared with SLT alone.

Is neuromuscular electrical stimulation (NMES) plus SLT more effective than sham NMES plus SLT?

Four studies analysed NMES (Baijens 2013; Heijnen 2012; Park 2018; Zhang 2025). The evidence is very uncertain about the effect of NMES plus SLT on swallowing efficiency and safety after four weeks of treatment.

Our certainty in these findings is very low due to imprecision (small sample size), high risk of bias related to deviations from intended interventions, measurement of the outcome, and the randomisation process, and poor reporting.

Application of the evidence

The evidence is very uncertain, making it difficult to address any clinical suggestions. We can hypothesise that the clinical applicability of expiratory muscle strength training (EMST) interventions appears to be linked to how EMST is provided and how devices are used in this context. Of the three studies that investigated EMST, all three delivered it as a home-based intervention, with a blinded speech-language therapist providing initial training and monitoring participants' performance. Based on this setup, it can be inferred that implementing EMST in a real-world setting is feasible. Nevertheless, further research of better methodological quality is needed to determine the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson's disease.

Limitations of the evidence included in the review

Assessing the certainty of the evidence was challenging due to poor, incomplete, or minimal information reporting. We deemed only two studies included in the analysis to have a low risk of bias across all domains. We downgraded the certainty of the evidence due to imprecision (small sample size) and to a high risk of bias in the randomisation process, missing outcome data, and measurement of the outcome. Given the variability in the risk of bias across multiple studies and the overall very low certainty of the evidence, the results of the data analysis should be interpreted with caution.

Completeness of published studies

We considered the reporting of most included studies to be poor or incomplete. Some studies simply reported the number of included participants, implying no attrition. However, we suspect that dropouts may have been excluded from the reported results in some studies, thereby introducing bias. Some did not state the methods used for randomisation or blinding of personnel and participants.

We contacted the authors of all studies with missing data or requiring additional information, but only a few responded. The authors of two studies informed us that they could not share their datasets, as the studies were conducted many years ago and they no longer held the necessary permissions or licences to access the data (Baijens 2013; Logemann 2008). Thus, the completeness of study reporting was limited, leading to many studies that could not contribute data to the analysis.

Finally, one of the key limitations of the included studies was the small sample size. This remains a common challenge in rehabilitation RCTs, often resulting in limited statistical power, reduced generalisability of findings, and lower confidence in the evidence, as reflected in the GRADE assessments. Addressing this issue in future research is essential to strengthening the evidence base and providing more robust conclusions [181, 182, 183].

Completeness of population descriptions

Many included studies were either unclear about or did not report key clinical details of participants' condition, including time since the onset of Parkinson's disease. The severity of the disease was most commonly assessed using the Hoehn and Yahr (H&Y) Scale, which primarily evaluates motor symptoms and disease progression (Baijens 2013; Claus 2021; Dashtelei 2024; Heijnen 2012; Khedr 2019; Logemann 2008; Manor 2013; Mohseni 2023; Park 2018; Plaza 2022; Riboldazzi 2020; Troche 2010; Troche 2023; Xie 2018). In contrast, only four studies used the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) (Cebi 2025; Claus 2021; Mohseni 2023; Troche 2023). The MDS-UPDRS offers a more comprehensive assessment, encompassing motor and non-motor symptoms, including swallowing, saliva control, and speech functions. To assess the severity and stage of Parkinson's disease, it is recommended that both the MDS-UPDRS and H&Y Scale are reported [98, 184, 185]. Given that the severity of dysphagia in individuals with Parkinson’s disease does not always correlate directly with motor symptoms, it is also crucial to document the specific phenotype of dysphagia at baseline, including Parkinsonian symptoms of swallowing (e.g. tongue tremors, lingual pumping, freezing and festination), as well as the phase of swallowing impairments [31]. In recent years, increasing evidence has underscored the significant influence of cognitive impairments on swallowing function in individuals with Parkinson’s disease [186, 187]. However, this critical factor was seldom reported in the included studies. To enhance the quality and relevance of future research, cognitive status should be systematically assessed and reported as part of the baseline assessment.

Completeness of outcome data

Significant heterogeneity in the interventions and comparators among the included studies limited the possibility of pooling results. Consequently, we performed only one meta-analysis.

The included studies also used a variety of methods and instruments for assessing the same outcomes. For instance, seven of nine studies that assessed the critical outcome of swallowing safety used the Penetration-Aspiration Scale (PAS) [47] (Baijens

2013; Cebi 2025; Claus 2021; Dashtelei 2024; Khedr 2019; Logemann 2008; ; Park 2018). Other studies used videofluoroscopy (VFS) or fiberoptic endoscopic evaluation of swallowing (FEES). Although all three methods are commonly used in swallowing assessment, they offer different perspectives on the potential entry of food into the airway, leading to varying interpretations. Additionally, the PAS reflects two distinct severity categories: penetration (scores 2–5) and aspiration (scores 6–8). Hence, a change from a score of 6 to 5 may hold greater clinical relevance – marking a shift from aspiration to penetration – than a change from 8 to 6, even though the latter reflects a broader numerical improvement. This nuance highlights a limitation of the scale in capturing clinically meaningful progress or deterioration. Furthermore, there were discrepancies in the food administration protocols: while some studies tested swallowing safety using multiple consistencies, such as liquids, semi-solids, and solids, others used only one consistency, most commonly semi-solid. This variability further contributed to the heterogeneity, leading to challenges in analysing and synthesising results. In this regard, the use of the International Dysphagia Diet Standardisation Initiative (IDDSI) framework is essential to minimise variability and reduce potential bias across studies [55, 188].

We encountered similar challenges for the outcome of swallowing efficiency. Several authors used non-validated assessment scales, while others modified existing validated scales. This lack of standardisation in measurement tools created further variability across studies, making comparing results and drawing consistent conclusions difficult. In addition, only a small proportion of the included studies reported follow-up assessments (Claus 2021; Dashtelei 2024; Logemann 2008; Manor 2013; Mohseni 2023; Xie 2018), and when performed, the timing of these evaluations varied considerably. Given that Parkinson's disease is a progressive neurodegenerative condition, incorporating consistent and well-defined follow-up assessments should be a standard component of study design to evaluate the long-term effectiveness and sustainability of rehabilitation interventions accurately.

To address heterogeneity in outcome measures, a multidisciplinary steering committee led by experts developed the Core Outcome Set for Dysphagia Interventions in Parkinson's Disease (COS-DIP). The COS-DIP supports the systematic measurement and reporting of a core set of outcomes across all clinical trials [189].

Completeness of intervention descriptions

While the included studies described interventions in their respective methods sections, these descriptions were often insufficient to enable confident, precise replication, particularly for usual care or traditional treatment (Baijens 2013; Heijnen 2012; Manor 2013; Park 2018; Riboldazzi 2020). For instance, some authors referred vaguely to "traditional speech and language therapy" without further detail. In contrast, others included in their definition of "usual care" an extensive list of swallowing exercises targeting strength, sensory stimulation, or compensatory manoeuvres. These treatments encompassed various swallowing rehabilitation approaches, each targeting different swallowing outcomes, and therefore should not be broadly classified under the term "usual care" or "traditional treatment." Another key issue was the incomplete reporting of the number of intervention sessions and dosage across studies, which made comparisons challenging and introduced potential bias. Moreover, there was significant variation in treatment frequency, ranging from a single session to daily interventions extending over a year. Although treatment fidelity is widely recognised as a crucial component of rehabilitation interventions in people with Parkinson's disease and dysphagia [190, 191], it was not consistently reported across all studies.

Medication intake also plays a crucial role in the management of people with Parkinson's disease. However, only a few studies reported whether the intervention was conducted during the "on" or "off" medication state, nor did they specify the type of medication used. Recent research has highlighted that dysphagia can significantly affect medication intake [17, 192]. Therefore, detailed reporting of medication status and type should be considered mandatory in intervention descriptions, as these factors can directly influence the outcomes of swallowing rehabilitation.

Finally, given that Parkinson's disease is a complex movement disorder, most individuals receive physiotherapy, occupational therapy, and neuropsychological support alongside other treatments. However, none of the included studies reported whether participants received these additional therapies. This information is fundamental, as such combined interventions could significantly influence the outcomes of swallowing rehabilitation. Current rehabilitation guidelines for Parkinson's disease emphasise the importance of a multimodal, multi-professional approach [193, 194]. Therefore, these factors should be systematically reported and considered when describing interventions to ensure a comprehensive understanding of all elements impacting swallowing rehabilitation outcomes [181, 182, 183].

Limitations of the review processes

We acknowledge the possibility of missing some relevant studies. To minimise selection bias, we implemented a comprehensive search strategy across multiple databases. We took measures to minimise biases in the review process. Working independently, two review authors (IB, MJDF) screened the titles and abstracts of identified studies using Covidence. We reviewed the references of relevant systematic reviews to identify any overlooked studies. To further reduce bias in data collection, selection, and extraction, at least two review authors independently performed these steps using Covidence.

Agreements and disagreements with other studies or reviews

Despite the increase in evidence since the publication of the previous Cochrane review by Deane and colleagues in 2001 [84], no definitive conclusion can be reached due to the very low certainty of the included evidence. Other systematic reviews have explored swallowing rehabilitation in individuals with Parkinson's disease and have offered clinical guidance [10, 12, 42, 48, 53], but the evidence is consistently of poor quality [12, 42]. Currently, there is insufficient evidence to support the effectiveness of rehabilitation interventions for oropharyngeal dysphagia in people with Parkinson's disease. Further research of better methodological quality is urgently needed to inform clinical decision-making in this field.

AUTHORS' CONCLUSIONS

Implications for practice

Although the number of randomised controlled trials exploring swallowing rehabilitation interventions in individuals with Parkinson's disease has grown since the publication of a 2001 Cochrane review by Deane and colleagues [84], uncertainty remains regarding their effectiveness. Among behavioural swallowing rehabilitation interventions, expiratory muscle strength training (EMST) appears to improve swallowing safety and efficiency, thereby improving swallowing status in individuals with Parkinson's disease, but the evidence is very uncertain. At the same time, these effects do not seem to be sustained in the long term. In general, the studies included in this review were predominantly small-scale and varied in methodological quality, thus resulting in very uncertain findings and constraining the robustness of the conclusions.

Implications for research

There is a clear need for robust, well-designed studies to advance the evidence base for swallowing rehabilitation in individuals with Parkinson's disease. Future randomised controlled trials should follow the updated CONSORT 2025 guidelines to ensure transparent and high-quality reporting [195]. Interventions must be clearly described, along with a well-defined comparator, including a precise explanation of what constitutes 'usual care'. Ensuring fidelity to the intervention is essential and can be supported by the use of the Template for Intervention Description and Replication (TIDieR) checklist [196]. Baseline assessments should include key Parkinson's disease-specific information, such as the Hoehn and Yahr stage, the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) scores, levodopa intake, dysphagia phenotype, and any cognitive impairments. Finally, outcome measures should reflect the Core Outcome Set for Dysphagia Interventions in Parkinson's Disease (COS-DIP) [189], and special attention should be given to blinding outcome assessors to reduce bias and strengthen the validity of the findings.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015816.pub2.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this review:

- Sign-off Editor (final editorial decision): Maria Gabriella Ceravolo, Department of Experimental and Clinical Medicine, Marche Polytechnic University, Ancona, Italy;
- Managing Editor (collated peer-reviewer comments, provided editorial guidance to authors, and edited the article): Marwah Anas El-Wegoud, Cochrane Central Editorial Service;
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, and supported the editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
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Contributions of authors

Irene Battel developed and co-ordinated the review. She contributed by writing the background, objectives, clinical interpretation of the data, and discussion. She approved the final manuscript.

Chiara Arienti contributed to the review by writing the methods, analysis, and interpreting the data. She approved the final manuscript.

Matteo Johann Del Furia contributed to the review by searching electronic databases and conference proceedings, screening titles and abstracts of references identified by the search, selecting and assessing trials, extracting trial and outcome data, and contributing to and approving the final manuscript.

Stefano Giuseppe Lazzarini contributed to the review by writing the methods and search strategy, and performing data collection, analysis, and interpretation of the results. He approved the final manuscript.

Tobias Warnecke contributed intellectually to the review by writing the discussion and interpreting the results. He approved the final manuscript.

Margaret Walshe died in March 2025. She contributed intellectually to the review by writing the background, objectives, and methods, and supervising the first part of the results section.

The living authors did not make substantive changes to those parts of the review to which Margaret contributed.

Declarations of interest

Irene Battel declares no conflicts of interest.

Chiara Arienti declares no conflicts of interest.

Matteo Johann Del Furia declares no conflicts of interest.

Stefano Giuseppe Lazzarini declares no conflicts of interest.

Tobias Warnecke received speaker fees/honoraria from Bial, Abbvie, Merz, Desitin, Pfizer, Britannia, Zambon, Neuraxpharm, Esteve, and Licher. He also received payment for travel, accommodation, subsistence, and conference registration from Abbvie. He received payment for expert advice from Stadapharm, Bial, Merz, Abbvie, and Phagenesis. He received funding for research from Abbvie and Ever Pharma. He is also a co-author of two included studies (Cebi 2025; Claus 2021).

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download by users of the Cochrane Library (see Supplementary material 1: full search strategies for each database; Supplementary material 2: full citation of each unique report for all included studies in the final review; Supplementary material 3: full citation of each unique report for all excluded studies at the full text screen in the final review; Supplementary material 4: full citation of each unique report for all studies awaiting classification in the final review; Supplementary material 5 : full citation of each unique report for all ongoing studies in the final review; Supplementary material 6: risk of bias assessments; Supplementary material 7: analysis data, including overall estimates and settings, subgroup estimates, and individual data rows; and Supplementary material 8: data package). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Covidence and detailed risk of bias assessment data with consensus responses to the signalling questions in the RoB 2 Excel tool are available from the authors on reasonable request.

REFERENCES

1. Balestrino R, Schapira AH. Parkinson disease. *European Journal of Neurology* 2020;**27**(1):27-42.

2. Ball N, Teo WP, Chandra S, Chapman J. Parkinson's disease and the environment. *Frontiers in Neurology* 2019;**10**:218.

3. Hirsch MA, Iyer SS, Sanjak M. Exercise-induced neuroplasticity in human Parkinson's disease: what is the evidence telling us? *Parkinsonism & Related Disorders* 2016;**22 Suppl 1**:S78-81.

4. Miller N. Swallowing in Parkinson's disease: clinical issues and management. *Neurodegenerative Disease Management* 2017;**7**(3):205-17.

5. Suttrup I, Warnecke T. Dysphagia in Parkinson's Disease. *Dysphagia* 2016;**31**(1):24-32.

6. Kalf JG, de Swart BJ, Bloem BR, Munneke M. Prevalence of oropharyngeal dysphagia in Parkinson's disease: a meta-analysis. *Parkinsonism & Related Disorders* 2012;**18**(4):311-5.

7. Dilmaghani S, Atieh J, Khanna L, Hosfield EA, Camilleri M, Katzka DA. Severity of dysphagia is associated with hospitalizations and mortality in patients with Parkinson's disease. *Neurogastroenterology and Motility* 2022;**34**(6):e14280.

8. Hobson P, Meara J. Mortality and quality of death certification in a cohort of patients with Parkinson's disease and matched controls in North Wales, UK at 18 years: a community-based cohort study. *BMJ Open* 2018;**8**(2):e018969.

9. Plowman-Prine EK, Sapienza CM, Okun MS, Pollock SL, Jacobson C, Wu SS, et al. The relationship between quality of life and swallowing in Parkinson's disease. *Movement Disorders* 2009;**24**(9):1352-8.

10. Battel I, Calvo I, Walshe M. Interventions involving biofeedback to improve swallowing in people with Parkinson disease and dysphagia: a systematic review. *Archives of Physical Medicine and Rehabilitation* 2020;**102**(2):314-22.

11. Chang MC, Park JS, Lee BJ, Park D. Effectiveness of pharmacologic treatment for dysphagia in Parkinson's disease: a narrative review. *Neurological Sciences* 2021;**42**(2):513-9.

12. Cheng I, Sasegbon A, Hamdy S. Dysphagia treatments in Parkinson's disease: a systematic review and meta-analysis. *Neurogastroenterology & Motility* 2023;**35**(8):e14517.

13. Akbar U, Dham B, He Y, Hack N, Wu S, Troche M, et al. Incidence and mortality trends of aspiration pneumonia in Parkinson's disease in the United States, 1979-2010. *Parkinsonism & Related Disorders* 2015;**21**(9):1082-6.

14. Rinaldi D, Imbalzano G, Galli S, Bianchini E, Ledda C, De Carolis L, et al. The impact of dysphagia in Parkinson's disease patients treated with levodopa/carbidopa intestinal gel. *Parkinsonism & Related Disorders* 2023;**109**:105368.

15. Fabbri M, Coelho M, Abreu D, Guedes LC, Rosa MM, Godinho C, et al. Dysphagia predicts poor outcome in late-stage Parkinson's disease. *Parkinsonism & Related Disorders* 2019;**64**:73-81.

16. Müller J, Wenning GK, Verny M, McKee A, Chaudhuri KR, Jellinger K, et al. Progression of dysarthria and dysphagia in postmortem-confirmed parkinsonian disorders. *Archives of Neurology* 2001;**58**(2):259-64.

17. Buhmann C, Bihler M, Emich K, Hidding U, Potter-Nerger M, Gerloff C, et al. Pill swallowing in Parkinson's disease: a prospective study based on flexible endoscopic evaluation of swallowing. *Parkinsonism & Related Disorders* 2019;**62**:51-6.

18. Chaudhuri KR, Healy DG, Schapira AH. Non-motor symptoms of Parkinson's disease: diagnosis and management. *Lancet Neurology* 2006;**5**(3):235-45.

19. Hunter PC, Crameri J, Austin S, Woodward MC, Hughes AJ. Response of parkinsonian swallowing dysfunction to dopaminergic stimulation. *Journal of Neurology, Neurosurgery and Psychiatry* 1997;**63**(5):579-83.

20. Mandell LA, Niederman MS. Aspiration pneumonia. *New England Journal of Medicine* 2019;**380**(7):651-63.

21. Nienstedt JC, Bihler M, Niessen A, Plaetke R, Potter-Nerger M, Gerloff C, et al. Predictive clinical factors for penetration and aspiration in Parkinson's disease. *Neurogastroenterology & Motility Journal* 2019;**31**(3):e13524.

22. Won JH, Byun SJ, Oh BM, Park SJ, Seo HG. Risk and mortality of aspiration pneumonia in Parkinson's disease: a nationwide database study. *Scientific Reports* 2021;**11**(1):6597.

23. Hegland KW, Okun MS, Troche MS. Sequential voluntary cough and aspiration or aspiration risk in Parkinson's disease. *Lung* 2014;**192**(4):601-8.

24. Pitts T, Troche M, Mann G, Rosenbek J, Okun MS, Sapienza C. Using voluntary cough to detect penetration and aspiration during oropharyngeal swallowing in patients with Parkinson disease. *Chest* 2010;**138**(6):1426-31.

25. Troche MS, Okun MS, Rosenbek JC, Musson N, Fernandez HH, Rodriguez R, et al. Aspiration and swallowing in Parkinson disease and rehabilitation with EMST: a randomized trial. *Neurology* 2010;**75**(21):1912-9.

26. Troche MS, Schumann B, Brandimore AE, Okun MS, Hegland KW. Reflex cough and disease duration as predictors of swallowing dysfunction in Parkinson's disease. *Dysphagia* 2016;**31**(6):757-64.

27. Wheeler HK, Troche MS, Brandimore AE, Davenport PW, Okun MS. Comparison of voluntary and reflex cough effectiveness in Parkinson's disease. *Parkinsonism & Related Disorders* 2014;**20**(11):1226-30.

28. Simons JA. Swallowing dysfunctions in Parkinson's disease. *International Review of Neurobiology* 2017;**134**:1207-38.

29. Pflug C, Bihler M, Emich K, Niessen A, Nienstedt JC, Flugel T, et al. Critical dysphagia is common in Parkinson disease and occurs even in early stages: a prospective cohort study. *Dysphagia* 2018;**33**(1):41-50.

30. Troche MS, Sapienza CM, Rosenbek JC. Effects of bolus consistency on timing and safety of swallow in patients with Parkinson's disease. *Dysphagia* 2008;**23**(1):26-32.

31. Labeit B, Claus I, Muhle P, Lapa S, Suntrup-Krueger S. Oropharyngeal freezing and its relation to dysphagia - an analogy to freezing of gait. *Parkinsonism Related Disorders* 2020;**75**:1-6.

32. Fukuoka T, Ono T, Hori K, Wada Y, Uchiyama Y, Kasama S, et al. Tongue pressure measurement and videofluoroscopic study of swallowing in patients with Parkinson's disease. *Dysphagia* 2019;**34**(1):80-8.

33. Curtis JA, Molfenter S, Troche MS. Predictors of residue and airway invasion in Parkinson's disease. *Dysphagia* 2020;**35**(2):220-30.

34. Curtis JA, Molfenter SM, Troche MS. Pharyngeal area changes in Parkinson's disease and its effect on swallowing safety, efficiency, and kinematics. *Dysphagia* 2020;**35**(2):389-98.

35. Warnecke T, Suttrup I, Schroder JB, Osada N, Oelenberg S, Hamacher C, et al. Levodopa responsiveness of dysphagia in advanced Parkinson's disease and reliability testing of the FEES-Levodopa-test. *Parkinsonism & Related Disorders* 2016;**28**:100-6.

36. Warnecke T, Labeit B, Schroeder J, Reckels A, Ahring S, Lapa S, et al. Neurogenic dysphagia: a systematic review and proposal of a classification system. *Neurology* 2020;**96**(6):e876-89.

37. Battel I, Walshe M. An intensive neurorehabilitation programme with sEMG biofeedback to improve swallowing in idiopathic Parkinson's disease (IPD): a feasibility study. *International Journal of Language & Communication Disorders* 2022;**58**(3):813-25.

38. Pizzorni N, Ciammola A, Casazza G, Ginocchio D, Bianchi F, Feroldi S, et al. Predictors of malnutrition risk in neurodegenerative diseases: the role of swallowing function. *European Journal of Neurology* 2022;**29**(8):2493-8.

39. Athukorala RP, Jones RD, Sella O, Huckabee ML. Skill training for swallowing rehabilitation in patients with Parkinson's disease. *Archives of Physical Medicine and Rehabilitation* 2014;**95**(7):1374-82.

40. Manor Y, Mootanah R, Freud D, Giladi N, Cohen JT. Video-assisted swallowing therapy for patients with Parkinson's disease. *Parkinsonism & Related Disorders* 2013;**19**(2):207-11.

41. Van Hooren MR, Baijens LW, Voskuilen S, Oosterloo M, Kremer B. Treatment effects for dysphagia in Parkinson's disease: a systematic review. *Parkinsonism & Related Disorders* 2014;**20**(8):800-7.

42. Hirschwald J, Hofacker J, Duncan S, Walshe M. Swallowing outcomes in dysphagia interventions in Parkinson's disease: a scoping review. *BMJ Evidence-Based Medicine* 2022;**28**(2):111-8. [DOI: 10.1136/bmjebm-2022-112082]

43. Perry SE, Borders JC, Dakin AE, Troche MS. Characterizing quality of life in caregivers of people with Parkinson's disease and dysphagia. *Dysphagia* 2022;**37**(3):523-32.

44. Attrill S, White S, Murray J, Hammond S, Doeltgen S. Impact of oropharyngeal dysphagia on healthcare cost and length of stay in hospital: a systematic review. *BMC Health Services Research* 2018;**18**(1):594.

45. Marin S, Serra-Prat M, Ortega O, Audouard Fericgla M, Valls J, Palomera E, et al. Healthcare costs of post-stroke oropharyngeal dysphagia and its complications: malnutrition and respiratory infections. *European Journal of Neurology* 2021;**28**(11):3670-81.

46. Di Luca DG, McArthur EW, Willis A, Martino R, Marras C. Clinical and economic outcomes associated with dysphagia in hospitalized patients with Parkinson's disease. *Journal of Parkinson's Disease* 2021;**11**(4):1965-71.

47. Rosenbek JC, Robbins JA, Roecker EB, Coyle JL, Wood JL. A penetration-aspiration scale. *Dysphagia* 1996;**11**(2):93-8.

48. Gandhi P, Steele CM. Effectiveness of interventions for dysphagia in Parkinson disease: a systematic review. *American Journal of Speech-Language Pathology* 2022;**31**(1):463-85.

49. Menezes C, Melo A. Does levodopa improve swallowing dysfunction in Parkinson's disease patients? *Journal of Clinical Pharmacy and Therapeutics* 2009;**34**(6):673-6.

50. Warnecke T, Hamacher C, Oelenberg S, Dziewas R. Off and on state assessment of swallowing function in Parkinson's disease. *Parkinsonism & Related Disorders* 2014;**20**(9):1033-4.

51. Craig P, Dieppe P, Macintyre S, Michie S, Nazareth I, Petticrew M. Developing and evaluating complex interventions: the new Medical Research Council guidance. *BMJ* 2008;**337**:a1655. [DOI: 10.1136/bmj.a1655]

52. Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. *BMJ* 2021;**374**:n2061. [DOI: 10.1136/bmj.n2061]

53. Kim JY, Kim HH. Effects of behavioural swallowing therapy in patients with Parkinson's disease: a systematic review. *International Journal of Speech-Language Pathology* 2023;**25**(2):269-80.

54. Winiker K, Kertscher B. Behavioural interventions for swallowing in subjects with Parkinson's disease: a mixed methods systematic review. *International Journal of Language & Communication Disorders* 2023;**58**(4):1375-404. [DOI: 10.1111/1460-6984.12865] [PMID: 36951546]

55. Cichero JAY. Evaluating chewing function: expanding the dysphagia field using food oral processing and the IDDSI framework. *Journal of Texture Studies* 2020;**51**(1):56-66.

56. Da Silva N, Verri E, Palinkas M, Hallak J, Regalo S, Siessere S. Impact of Parkinson's disease on the efficiency of masticatory cycles: electromyographic analysis. *Medicina Oral, Patología Oral y Cirugía Bucal* 2019;**24**(3):e314-18.

57. Matsuo K, Fujishima I. Textural changes by mastication and proper food texture for patients with oropharyngeal dysphagia. *Nutrients* 2020;**12**(6):1613.

58. Steele M, Alsanei WA, Ayanikalath S, Barbon CE, Chen J, Cichero JA, et al. The influence of food texture and liquid consistency modification on swallowing physiology and function: a systematic review. *Dysphagia* 2015;**30**(1):2-26.

59. Flynn E, Smith CH, Walsh CD, Walshe M. Modifying the consistency of food and fluids for swallowing difficulties in dementia. *Cochrane Database of Systematic Reviews* 2018, Issue 9. Art. No: CD011077. [DOI: 10.1002/14651858.CD011077.pub2]

60. Logemann JA, Gensler G, Robbins J, Lindblad AS, Brandt D, Hind JA, et al. A randomized study of three interventions for aspiration of thin liquids in patients with dementia or Parkinson's disease. *Journal of Speech, Language, and Hearing Research* 2008;**51**(1):173-83.

61. Newman R, Vilardell N, Clavé P, Speyer R. Effect of bolus viscosity on the safety and efficacy of swallowing and the kinematics of the swallow response in patients with oropharyngeal dysphagia: white paper by the European Society for Swallowing Disorders (ESSD). *Dysphagia* 2016;**31**(2):232-49.

62. Robbins J, Gensler G, Hind J, Logemann JA, Lindblad AS, Brandt D, et al. Comparison of 2 interventions for liquid aspiration on pneumonia incidence: a randomized trial. *Annals of Internal Medicine* 2008;**148**(7):509-18.

63. Langmore SE, Pisegna JM. Efficacy of exercises to rehabilitate dysphagia: a critique of the literature. *International Journal of Speech-Language Pathology* 2015;**17**(3):222-9.

64. Robbins J, Butler SG, Daniels SK, Diez GR, Langmore S, Lazarus CL, et al. Swallowing and dysphagia rehabilitation: translating principles of neural plasticity into clinically oriented evidence. *Journal of Speech, Language, and Hearing Research* 2008;**51**(1):S276-300.

65. Cocks N, Rafols J, Embley E, Hill K. Expiratory muscle strength training for drooling in adults with Parkinson's disease. *Dysphagia* 2022;**37**(6):1525-31.

66. Reyes A, Castillo A, Castillo J, Cornejo I. The effects of respiratory muscle training on peak cough flow in patients with Parkinson's disease: a randomized controlled study. *Clinical Rehabilitation* 2018;**32**(10):1317-27.

67. Jenks J, Pitts LL. Effects of an intensive exercise-based swallowing program for persons with Parkinson's disease and complex medical history: a single-case experiment. *American Journal of Speech-Language Pathology* 2019;**28**(3):1268-74.

68. Hegland KW, Troche M, Brandimore A. Relationship between respiratory sensory perception, speech, and swallow in Parkinson's disease. *Movement Disorders Clinical Practice* 2019;**6**(3):243-9.

69. Miles A, Jardine M, Johnston F, de Lisle M, Friary P, Allen J. Effect of Lee Silverman Voice Treatment (LSVT LOUD(R)) on swallowing and cough in Parkinson's disease: a pilot study. *Journal of the Neurological Sciences* 2017;**383**:180-7.

70. Martin RE. Neuroplasticity and swallowing. *Dysphagia* 2009;**24**(2):218-29.

71. Huckabee ML, Burnip E. Still rethinking rehab: motor learning treatment approaches for dysphagia. *Perspectives of the ASHA Special Interest Groups* 2018;**3**(13):146-56.

72. Troche MS, Brandimore AE, Foote KD, Morishita T, Chen D, Hegland KW, et al. Swallowing outcomes following unilateral STN vs. GPi surgery: a retrospective analysis. *Dysphagia* 2014;**29**(4):425-31.

73. Khedr EM, Mohamed KO, Soliman RK, Hassan AM, Rothwell JC. The effect of high-frequency repetitive transcranial magnetic stimulation on advancing Parkinson's disease with dysphagia: double blind randomized clinical trial. *Neurorehabilitation and Neural Repair* 2019;**33**(6):442-52.

74. Sasegbon A, Hammerbeck U, Michou E, Cheng I, Zhang M, James C, et al. A feasibility pilot study of the effects of neurostimulation on swallowing function in Parkinson's Disease. *Association of Medical Research Charities* 2022;**3**:19.

75. Park JS, Oh DH, Hwang NK, Lee JH. Effects of neuromuscular electrical stimulation in patients with Parkinson's disease and dysphagia: a randomized, single-blind, placebo-controlled trial. *NeuroRehabilitation* 2018;**42**(4):457-63.

76. Logemann J. Approaches to management of disordered swallowing. *Clinical Gastroenterology* 1991;**5**:269-80.

77. Regan J, Walshe M, Tobin WO. Immediate effects of thermal-tactile stimulation on timing of swallow in idiopathic Parkinson's disease. *Dysphagia* 2010;**25**(3):207-15.

78. Krival K, Pelletier C, Kelchner L. Effects of carbonate vs thin and thickened liquids on swallowing in adults with stroke. *Dysphagia* 2008;**23**:428.

79. Price K, Isbister G, Long S, Mirams J, Smithard D. Are bubbles the future of dysphagia rehabilitation: a systematic review analysing evidence on the use of carbonated liquids in dysphagia rehabilitation. *Geriatrics* 2023;**8**(1):6.

80. Yang CW, Chen RD, Feng MT, Zhang MZ, Liu W, Liu XC, et al. The therapeutic effect of capsaicin on oropharyngeal dysphagia: a systematic review and meta-analysis. *Frontiers in Aging Neuroscience* 2022;**14**:931016.

81. Pitts T, Bolser D, Rosenbek J, Troche M, Okun MS, Sapienza C. Impact of expiratory muscle strength training on voluntary cough and swallow function in Parkinson disease. *Chest* 2009;**135**(5):1301-8.

82. McNaney R, Miller N, Vines J, Olivier P, Ladha K, Jackson D, et al. The feasibility and acceptability of using a novel wrist worn cueing device to self-manage drooling problems in people with Parkinson's disease: a pilot study. *Journal of Rehabilitation and Assistive Technologies Engineering* 2019;**6**:2055668319852529.

83. El Sharkawi A, Ramig L, Logemann JA, Pauloski BR, Rademaker AW, Smith CH, et al. Swallowing and voice effects of

Lee Silverman Voice Treatment (LSVT): a pilot study. *Journal of Neurology, Neurosurgery and Psychiatry* 2002;**72**(1):31-6.

84. Deane KH, Whurr R, Clarke CE, Playford ED, Ben-Shlomo Y. Non-pharmacological therapies for dysphagia in Parkinson's disease. *Cochrane Database of Systematic Reviews* 2001, Issue 1. Art. No: CD002816. [DOI: 10.1002/14651858.CD002816]

85. Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill. Methodological Expectations of Cochrane Intervention Reviews. Cochrane: London, Version August 2023. Available from https://www.cochrane.org/authors/handbooks-and-manuals/mecir.

86. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: 10.1136/bmj.n71]

87. Berg D, Postuma RB, Adler CH, Bloem BR, Chan P, Dubois B, et al. MDS research criteria for prodromal Parkinson's disease. *Movement Disorders* 2015;**30**(12):1600-11.

88. Neubauer PD, Rademaker AW, Leder SB. The Yale Pharyngeal Residue Severity Rating Scale: an anatomically defined and image-based tool. *Dysphagia* 2015;**30**(5):521-8.

89. Rommel N, Borgers C, Van BD, Goeleven A, Dejaeger E, Omari TI. Bolus Residue Scale: an easy-to-use and reliable videofluoroscopic analysis tool to score bolus residue in patients with dysphagia. *International Journal of Otolaryngology* 2015;**178**(2):393-8.

90. Eisenhuber E, Schima W, Schober E, Pokieser P, Stadler A, Scharitzer M, et al. Videofluoroscopic assessment of patients with dysphagia: pharyngeal retention is a predictive factor for aspiration. *American Journal of Roentgenology* 2002;**178**(2):393-8.

91. O'Neil KH, Purdy M, Falk J, Gallo L. The Dysphagia Outcome and Severity Scale. *Dysphagia* 1999;**14**(3):139-45.

92. Tohara H, Palmer JB, Reynolds K, Kuhlemeier KV, Palmer S. Dysphagia severity scale. *Kokubyo Gakkai Zasshi* 2003;**70**:242-8.

93. Crary MA, Mann GD, Groher ME. Initial psychometric assessment of a functional oral intake scale for dysphagia in stroke patients. *Archives of Physical Medicine and Rehabilitation* 2005;**86**(8):1516-20.

94. McHorney CA, Robbins J, Lomax K, Rosenbek JC, Chignell K, Kramer AE, et al. The SWAL-QOL and SWAL-CARE outcomes tool for oropharyngeal dysphagia in adults: III. Documentation of reliability and validity. *Dysphagia* 2002;**17**(2):97-114.

95. Stratton RJ, Hackston A, Longmore D, Dixon R, Price S, Stroud M, et al. Malnutrition in hospital outpatients and inpatients: prevalence, concurrent validity and ease of use of the 'malnutrition universal screening tool' ('MUST') for adults. *British Journal of Nutrition* 2004;**92**(5):799-808.

96. Covidence. Version accessed prior to 9 January 2026. Melbourne, Australia: Veritas Health Innovation, 2023. Available at https://www.covidence.org.

97. Hoehn MM, Yahr MD. Parkinsonism: onset, progression and mortality. *Neurology* 1967;**17**(5):427-42.

98. Goetz CG, Fahn S, Martinez-Martin P, Poewe W, Sampaio C, Stebbins GT, et al. Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): process, format, and clinimetric testing plan. *Movement Disorders* 2007;**22**(1):41-7.

99. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898.

100. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial (chapter last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

101. Cohen J. A power primer. *Psychological Bulletin* 1992;**112**(1):155-9. [DOI: 10.1037//0033-2909.112.1.155]

102. Higgins JP, Li T, Deeks JJ, editor(s). Chapter 6: Choosing effect measures and computing estimates of effect (chapter last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

103. Higgins JP, Eldridge S, Li T. Chapter 23: Including variants on randomized trials (chapter last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

104. Sterne JA, Sutton AJ, Ioannidis JP, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ* 2011;**343**:d4002.

105. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clinical Trials* 1986;**7**(3):177-88. [DOI: 10.1016/0197-2456(86)90046-2]

106. McKenzie JE, Brennan SE. Chapter 12: Synthesizing and presenting findings using other methods (chapter last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

107. Deeks JJ, Higgins JP, Altman DG, McKenzie JE, Veroniki AA, editor(s). Chapter 10: Analysing data and undertaking meta-analyses (chapter last updated November 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

108. Camerucci E, Stang CD, Hajeb M, Turcano P, Mullan AF, Martin P, et al. Early-onset Parkinsonism and early-onset Parkinson's disease: a population-based study (2010-2015). *Journal of Parkinson's Disease* 2021;**11**(3):1197-207.

109. Martinez-Ramirez D, Almeida L, Giugni JC, Ahmed B, Higuchi MA, Little CS, et al. Rate of aspiration pneumonia in hospitalized Parkinson's disease patients: a cross-sectional study. *BMC Neurology* 2015;**15**:104.

110. Review Manager (RevMan). The Cochrane Collaboration: 9.17.0, 2025. Available at revman.cochrane.org.

111. Negrini S, Selb M, Kiekens C, Todhunter-Brown A, Arienti C, Stucki G, et al. Rehabilitation methodology meeting participants. Rehabilitation definition for research purposes: a global stakeholders' initiative by Cochrane Rehabilitation. *American Journal of Physical Medicine & Rehabilitation* 2022;**101**(7):e100-7. [DOI: 10.1097/PHM.0000000000002031] [PMID: 35583514]

112. Arienti C, Armijo-Olivo S, Minozzi S, Tjosvold L, Lazzarini SG, Patrini M, et al. Methodological issues in rehabilitation research: a scoping review. *Archives of Physical Medicine and Rehabilitation* 2021;**102**(8):1614-22.e14. [DOI: 10.1016/j.apmr.2021.04.006] [PMID: 33989598]

113. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, Guyatt GH. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence (chapter last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

114. Schünemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions (chapter last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from training.cochrane.org/handbook.

115. Baijens LW, Speyer R, Passos VL, Pilz W, Van der Kruis J, Haarmans S, et al. Surface electrical stimulation in dysphagic Parkinson patients: a randomized clinical trial. *Laryngoscope* 2013;**123**(11):E38-44. [DOI: 10.1002/lary.24119]

116. Baijens LW, Speyer R, Passos VL, Pilz W, Van der Kruis J, Haarmans S, et al. Surface electrical stimulation in dysphagic Parkinson patients: a randomized clinical trial. *Dysphagia* 2014;**29**(3):404. [DOI: 10.1007/s00455-014-9536-7]

117. Cebi I, Graf LH, Schütt M, Hormozi M, Klocke P, Löffler M, et al. Oral transport, penetration, and aspiration in PD: insights from a RCT on STN + SNr stimulation. *Dysphagia* 2025;**40**(4):811-22. [DOI: 10.1007/s00455-024-10779-y]

118. Cebi I, Graf LH, Hormozi M, Schneider M, Loffler MA, Gharabaghi A, et al. Restitution of oral transport, penetration, and aspiration with nigral stimulation in Parkinson's disease? In: Clinical Neurophysiology. 2022:e40. [DOI: 10.1016/j.clinph.2022.01.075]

119. Schoellmann A, Stoll L, Hobl B, Meisner C, Gharabaghi A, Weiss D. Clinical trial protocol: restitution of oral transport, penetration and aspiration with nigral stimulation in Parkinson's disease? In: Movement Disorders. 2018:S158.

120. NCT03470324. Combined stimulation of STN and SNr for dysphagia in Parkinson's disease. clinicaltrials.gov/study/NCT03470324 (first posted 19 March 2018).

121. Claus I, Muhle P, Czechowski J, Ahring S, Labeit B, Suntrup-Krueger S, et al. Expiratory muscle strength training for therapy of pharyngeal dysphagia in Parkinson's disease. *Movement Disorders* 2021;**36**(8):1815-24. [DOI: 10.1002/mds.28552]

122. Claus I, Muhle P, Suttrup J, Schroeder JB, Suntrup-Krueger S, Dziewas R, et al. Expiratory muscle strength training (EMST) for treatment of pharyngeal dysphagia in Parkinson's disease: a randomized controlled trial. *Dysphagia* 2020;**35**(1):143. [DOI: 10.1007/s00455-019-10078-x]

123. NCT02461082. EMST and TMS for Treatment of Dysphagia in Parkinson's Disease [Expiratory muscle strength training (EMST) and transcranial magnetic stimulation (TMS) for treatment of swallowing disorders in Parkinson's Disease]. clinicaltrials.gov/study/NCT02461082 (first posted 29 May 2015).

124. Dashtelei AA, Nitsche MA, Salehinejad MA, Habibi AH, Bakhtyiari J, Khatoonabadi AR. Adjunctive transcranial direct current stimulation to improve swallowing functions in Parkinson's disease. *Experimental and Clinical Sciences Journal* 2024;**23**:95-107. [DOI: 10.17179/excli2023-6496]

125. Irct20190412043248N1. Effect of brain stimulation on the improvement of swallowing disorder in Parkinson's patients [The effectiveness of combined transcranial Direct Current Stimulation (tDCS) and conventional swallowing therapy on improvement of the swallowing disorder in persons with Parkinson's disease]. trialsearch.who.int/Trial2.aspx?TrialID=Irct20190412043248N1 (first posted 15 June 2019).

126. Feng Q, Cheng Y, Lie P, Zhou X, Wang Y, Chen R, et al. Effects of vocalization training on drooling severity in Parkinson's disease. *Zhongguo xiandai sheng jing jibing zashi [Chinese Journal of Contemporary Neurology & Neurosurgery]* 2019;**19**(11):891-6. [DOI: 10.3969/j.issn.1672*6731.2019.11.014]

127. Heijnen BJ, Speyer R, Baijens LW, Bogaardt HC. Neuromuscular electrical stimulation versus traditional therapy in patients with Parkinson's disease and oropharyngeal dysphagia: effects on quality of life. *Dysphagia* 2012;**27**(3):336-45. [DOI: 10.1007/s00455-011-9371-z]

128. Khan GS, Kiyani MN, Ijaz T, Yousaf T, Imran S. Effectiveness of biofeedback therapy versus traditional therapy on dysphagia: a randomized control trial. *Medical Forum Monthly* 2025;**36**(7):16-20. [DOI: 10.60110/medforum.360703]

129. NCT07026032. Effects of biofeedback exercises on severity of dysphagia in Parkinson's clients. clinicaltrials.gov/study/NCT07026032 (first posted 18 June 2025).

130. Khedr EM, Mohamed KO, Soliman RK, Hassan AM, Rothwell JC. The effect of high-frequency repetitive transcranial magnetic stimulation on advancing Parkinson's disease with dysphagia: double blind randomized clinical trial. *Neurorehabilitation and Neural Repair* 2019;**33**(6):442-52. [DOI: 10.1177/1545968319847968]

131. NCT03317509. The effect of high frequency rTMS on advancing Parkinson’s disease with dysphagia [The effect of high frequency repetitive transcranial magnetic stimulation on advancing Parkinson's disease with dysphagia]. clinicaltrials.gov/study/NCT03317509 (first posted 20 October 2017).

132. Logemann JA, Gensler G, Robbins J, Lindblad AS, Brandt D, Hind JA, et al. A randomized study of three interventions for aspiration of thin liquids in patients with dementia or Parkinson's disease. *Journal of Speech, Language, and Hearing Research* 2008;**51**(1):173-83. [DOI: 10.1044/1092-4388(2008/013)]

133. NCT00000362. Randomized study of two Interventions for liquid aspiration: short-term and long-term effects. clinicaltrials.gov/study/NCT00000362 (first posted 23 June 2005).

134. Robbins J, Gensler G, Hind J, Logemann JA, Lindblad AS, Brandt D, et al. Comparison of 2 interventions for liquid aspiration on pneumonia incidence: a randomized trial. *Annals of Internal Medicine* 2008;**148**(7):509-18. [DOI: 10.7326/0003-4819-148-7-200804010-00007]

135. Manor Y, Mootanah R, Freud D, Giladi N, Cohen JT. Video-assisted swallowing therapy for patients with Parkinson's disease. *Parkinsonism & Related Disorders* 2013;**19**(2):207-11. [DOI: 10.1016/j.parkreldis.2012.10.004]

136. Manor Y, Freud D, Mootanah R, Gurevich T, Giladi N, Jacob CT. Video-assisted swallowing therapy improved quality of life measures more than traditional swallowing therapy in patients with Parkinson's disease. *Movement Disorders* 2010;**25**:S427. [DOI: 10.1002/mds.23162]

137. Mohseni Z, Saffarian A, Mohamadi R, Abolghasemi J, Habibi SA. Effect of conventional speech therapy combined with music therapy on swallowing in patients with Parkinson's disease (telerehabilitation): a randomized-controlled trial. *Middle East Journal of Rehabilitation & Health Studies* 2023;**10**(1):1-10. [DOI: 10.5812/mejrh-131572]

138. Park JS, Oh DH, Hwang NK, Lee JH. Effects of neuromuscular electrical stimulation in patients with Parkinson's disease and dysphagia: a randomized, single-blind, placebo-controlled trial. *NeuroRehabilitation* 2018;**42**(4):457-63. [DOI: 10.3233/nre-172306]

139. KCT0002511. Effects of neuromuscular electrical stimulation in patients with parkinson’s disease and dysphagia. cris.nih.go.kr/cris/en/search/search_result_st01.jsp?seq=9144 (first posted 27 October 2017).

140. Plaza E, Ruviaro Busanello-Stella A. Effects of a tongue training program in Parkinson's disease: analysis of electrical activity and strength of suprahyoid muscles. *Journal of Electromyography and Kinesiology* 2022;**63**:102642. [DOI: 10.1016/j.jelekin.2022.102642]

141. Riboldazzi G, Spinazza G, Beccarelli L, Prato P, Grecchi B, D'Abrosca F, et al. Effectiveness of expiratory flow acceleration in patients with Parkinson's disease and swallowing deficiency: a preliminary study. *Clinical Neurology and Neurosurgery* 2020;**199**:106249. [DOI: 10.1016/j.clineuro.2020.106249]

142. ChiCTR2000029371. Effectiveness of the expiratory flow acceleration technique in the reduction of bronchial exacerbations in patients with Parkinson's disease and swallowing deficiency: a preliminary study. chictr.org.cn/showproj.html?proj=47874 (first posted 25 January 2020).

143. Troche MS, Okun MS, Rosenbek JC, Musson N, Fernandez HH, Rodriguez R, et al. Aspiration and swallowing in Parkinson disease and rehabilitation with EMST: a randomized trial. *Neurology* 2010;**75**(21):1912-9. [DOI: 10.1212/WNL.0b013e3181fef115]

144. Troche M, Rosenbek J, Okun M, Sapienza C. Swallowing and breathing related outcomes following detraining in EMST. In: Dysphagia. Vol. 26. 2011:443-4. [DOI: 10.1007/s00455-011-9345-1]

145. Troche M, Rosenbek JC, Okun MS, Sapienza CM. Detraining effects and EMST in Parkinson's disease. In: Neurodegenerative Diseases. Vol. 8. 2011:305-10. [DOI: 10.1159/000327701]

146. Troche MS, Rosenbek JC, Okun MS, Sapienza CM. Swallowing related quality of life and EMST in Parkinson's disease. In: Neurodegenerative Diseases. Vol. 8. 2011. [DOI: 10.1159/000327701]

147. Troche M, Okun M, Rosenbek J, Musson N, Sapienza C. Swallow outcomes following intervention with expiratory muscle strength training (EMST) in Parkinson's disease: results of a randomized clinical trial. In: Dysphagia. Vol. 24. 2009:455-6. [DOI: 10.1007/s00455-009-9233-0]

148. Sapienza C, Troche M, Silverman E, Rosenbek J, Musson N, Okun M. Strength training outcomes for airway protection in PD. In: Neurodegenerative Diseases. Vol. 8. 2011. [DOI: 10.1159/000327701]

149. NCT00843739. Expiratory muscle training in patients with Parkinson's Disease (EMST). clinicaltrials.gov/study/NCT00843739 (first posted 12 February 2009).

150. Sapienza CM, Troche M, Silverman EP, Rosenbek J, Musson N. Strength training outcomes for airway protection in PD. In: Movement Disorders. Vol. 27. 2012:S139. [DOI: 10.1002/mds.25051]

151. Troche MS, Curtis JA, Sevitz JS, Dakin AE, Perry SE, Borders JC, et al. Rehabilitating cough dysfunction in Parkinson's Disease: a randomized controlled trial. *Movement Disorders* 2023;**38**(2):201-11. [DOI: 10.1002/mds.29268]

152. Doruk C, Curtis JA, Dakin AE, Troche MS. Cough and swallowing therapy and their effects on vocal fold bowing and laryngeal lesions. *Laryngoscope* 2023;**134**:1127-32. [DOI: 10.1002/lary.30922]

153. NCT02927691. Novel management of airway protection in Parkinson's disease: a clinical trial. clinicaltrials.gov/study/NCT02927691 (first posted 13 September 2013).

154. Troche M, Curtis J, Dakin A, Sevitz J, Borders J, Perry S, et al. Sensorimotor training for airway protection (SMTAP) versus expiratory muscle strength training (EMST) to rehabilitate cough and swallowing in Parkinson's Disease: a clinical trial. In: Dysphagia. Vol. 38. 2023:S90. [DOI: 10.1007/s00455-022-10456-y]

155. Xie T, Bloom L, Padmanaban M, Bertacchi B, Kang W, MacCracken E, et al. Long-term effect of low frequency stimulation of STN on dysphagia, freezing of gait and other motor symptoms in PD. *Journal of Neurology, Neurosurgery, and Psychiatry* 2018;**89**(9):989-94. [DOI: 10.1136/jnnp-2018-318060]

156. Xie T, Bloom L, Padmanaban M, Bertacchi B, Kang W, Dachman A, et al. Long-term effect of low frequency stimulation of STN on dysphagia, freezing of gait and other motor symptoms in PD. In: Movement Disorders. Vol. 33. 2018:S263.

157. Zhang B, He H, Zheng Y, Liu M, Wang L. Clinical study on high-frequency rTMS combined with motor imagery therapy for the treatment of Parkinson's disease with swallowing disorders. *Zhongguo Liaoyang Yixue [Chinese Journal of Convalescent Medicine]* 2025;**34**(3):53-6. [DOI: 10.13517/j.cnki.ccm.2025.03.011]

158. Manor Y, Giladi N, Cohen A, Fliss DM, Cohen JT. Validation of a swallowing disturbance questionnaire for detecting dysphagia in patients with Parkinson's disease. *Movement Disorder* 2007;**22**(13):1917-21.

159. Silbergleit AK, Schultz L, Jacobson BH, Beardsley T, Johnson AF. The Dysphagia Handicap Index: development and validation. *Dysphagia* 2012;**27**(1):46-52.

160. Jenkinson C, Fitzpatrick R, Peto V, Greenhall R, Hyman N. The Parkinson's Disease Questionnaire (PDQ-39): development and validation of a Parkinson's disease summary index score. *Age and Ageing* 1997;**26**(5):353-7.

161. NCT05700825. Rehabilitation of airway protection in Parkinson's Disease [Rehabilitation of airway protection in Parkinson's Disease: comparing in-person and telehealth service delivery models]. clinicaltrials.gov/study/NCT05700825 (first posted 26 January 2023).

162. NCT05837520. i-BiSSkApp for awallowing rehabilitation in Parkinson's Disease [At-home skill training using biofeedback for swallowing rehabilitation in Parkinson's disease]. clinicaltrials.gov/study/NCT05837520 (first posted 01 May 2023).

163. NCT06328296. Impact of Myofascial Release on Dysphagia in Parkinson's Disease Patients [A Randomized Controlled Study to Explore Effect of Myofascial Release on Dysphagia in Parkinson's Patients]. clinicaltrials.gov/study/NCT06328296 (first posted 25 March 2024).

164. NCT06370195. Effect of Stellate Ganglion Block on Drooling in Parkinson's Syndrome [Effect of Stellate Ganglion Block on Drooling in Parkinson's Syndrome: A Randomized Controlled Study]. clinicaltrials.gov/study/NCT06370195 (first posted 17 April 2024).

165. NCT06791590. Evaluating the Effects of Intermittent Oro-esophageal in Tracheotomy Patients With Neurogenic Dysphagia [Evaluating the Effects of Intermittent Oro-esophageal in Tracheotomy Patients With Neurogenic Dysphagia: a Randomized Study]. clinicaltrials.gov/study/NCT06791590 (first posted 24 January 2025).

166. NCT07153692. Clinical Efficacy and Mechanism of tDCS for Dysphagia in PD [Clinical Efficacy and Mechanism of tDCS for Dysphagia in Patients With Parkinson's Disease]. clinicaltrials.gov/study/NCT07153692 (first posted 04 September 2025).

167. ChiCTR2200060790. Clinical study on the effect of progressive inspiratory resistance training combined with swallowing exercise on swallowing function patients Parkinson's disease. chictr.org.cn/showproj.html?proj=159857 (first posted 11 June 2022).

168. ChiCTR2300076032. Application of percutaneous vagus nerve stimulation in dysphagia and rehabilitation of patients with Parkinson's disease. chictr.org.cn/showproj.html?proj=205707 (first posted 22 September 2023).

169. RBR-3pmvvnx. Application of an application connected to a game with stimulation of mastication and swallowing muscle activity in elderly people with Parkinson's disease. ensaiosclinicos.gov.br/rg/RBR-3pmvvnx (first posted 04 April 2023).

170. RBR-5ddt474. Effects of an intervention program in patients with swallowing difficulties and Parkinson's [Effects of an intervention program in dysphagic patients with dysphagia Parkinson's]. ensaiosclinicos.gov.br/rg/RBR-5ddt474 (first posted 17 November 2023).

171. Dismore L, Montague K, Carvalho L, Guerreiro T, Jackson D, Guan Y, Walker R. A protocol for the evaluation of a wearable device for monitoring of symptoms, and cueing for the management of drooling, in people with Parkinson's disease. *PLOS One* 2023;**18**(2):e0280727. [DOI: 10.1371/journal.pone.0280727]

172. ISRCTN59284050. A wearable device for people with Parkinson's disease to help with drooling and other symptoms of Parkinson's [A wearable device for cueing for the management of drooling, and monitoring of symptoms, in people with Parkinson's disease]. isrctn.com/ISRCTN59284050 (first posted 04 May 2022).

173. JPRN-UMIN000056111. Randomized Controlled Trial on the Clinical Effectiveness of Somato-Cognitive Coordination Therapy using mediVR KAGURA in Parkinson's Disease at Hoehn & Yahr Classification Stage III [Randomized Controlled Trial on the Clinical Effectiveness of Somato-Cognitive Coordination Therapy using mediVR KAGURA in Parkinson's Disease at Hoehn & Yahr Classification Stage III - RCT of Somato-Cognitive Coordination Therapy using mediVR KAGURA in Parkinson's Disease at Hoehn & Yahr Classification Stage III]. center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000064115 (first posted 09 November 2024).

174. Nakamori M, Matsuyama R, Toko M, Yamada H, Hayashi Y, Yoshikawa K, et al. Investigation of the improvement of swallowing function with low-frequency pulse waves or interference waves delivered via cervical electrical stimulation in patients with Parkinson's disease: A randomized controlled study protocol. *Contemporary Clinical Trials Communications* 2025;**44**:101463. [DOI: 10.1016/j.conctc.2025.101463]

175. JPRN-jRCTs062240041. A Randomized Controlled Trial of Cervical Electrical Stimulation for Parkinson's Disease [A Randomized Study Investigating the Improvement of Swallowing Function with Low-Frequency Pulse Waves and Interference Waves in Cervical Electrical Stimulation Rehabilitation for Parkinson's Disease]. jrct.mhlw.go.jp/latest-detail/jRCTs062240041 (first posted 29 July 2024).

176. Akbarkhodjaeva ZA, Mirzaeva KS. Effectiveness of a Multidisciplinary Rehabilitation Program in Improving Motor Function and Quality of Life in Parkinson's Disease Patients. In: Parkinsonism Relatated Disorders. Vol. 134. 2025:99. [DOI: 10.1016/j.parkreldis.2025.107571]

177. Nozaki S, Sugishita S, Imai T, Oguro D, Inokawa M, Matsui T, et al. Training for dysphagia with metronome improves swallowing function in parkinson disease. In: Dysphagia. Vol. 26. 2011:443. [DOI: 10.1007/s00455-011-9345-1]

178. Nozaki S, Matsui T, Yoshikawa H, Kaneto T, Domen K, Daimon T. Metronome therapy to treat dysphagia in patients with parkinson's disease. In: Dysphagia. Vol. 28. 2013:298. [DOI: 10.1007/s00455-013-9455-z]

179. ACTRN12618000090213. Expiratory Muscle Strength training (EMST) versus traditional dysphagia rehabilitation for intervention of dysphagia in Parkinson's disease in the community setting. anzctr.org.au/ACTRN12618000090213.aspx (first posted 19 January 2018).

180. NCT01420796. Swallowing and breathing exercises for dysphagia in Parkinson Disease [Effect of oral motor exercises and breathing exercises for dysphagia in Parkinson disease: randomized clinical trial]. clinicaltrials.gov/study/NCT01420796 (first posted 22 August 2011).

181. Côté P, Negrini S, Donzelli S, Kiekens C, Arienti C, Ceravolo MG, et al. Introduction to target trial emulation in rehabilitation: a systematic approach to emulate a randomized controlled trial using observational data. *European Journal of Physical and Rehabilitation Medicine* 2024;**60**(1):145-53. [DOI: 10.23736/S1973-9087.24.08435-1]

182. Levack WM, Gross DP, Martin RA, Every-Palmer S, Kiekens C, Cordani C, et al. Designing studies and reviews to produce informative, trustworthy evidence about complex interventions in rehabilitation: a narrative review and commentary. *European Journal of Physical and Rehabilitation Medicine* 2024;**60**(6):1088-99. [DOI: 10.23736/S1973-9087.24.08459-4]

183. Arienti C, Armijo-Olivo S, Ferriero G, Feys P, Hoogeboom T, Kiekens C, et al. The influence of bias in randomized controlled trials on rehabilitation intervention effect estimates: what we have learned from meta-epidemiological studies. *European Journal of Physical and Rehabilitation Medicine* 2024;**60**(1):135-44. [DOI: 10.23736/S1973-9087.23.08310-7]

184. Evers LJ, Krijthe JH, Meinders MJ, Bloem BR, Heskes TM. Measuring Parkinson's disease over time: the real-world within-subject reliability of the MDS-UPDRS. *Movement Disorders* 2019;**34**(10):1480-7. [DOI: 10.1002/mds.27790]

185. Skorvanek M, Martinez-Martin P, Kovacs N, Rodriguez-Violante M, Corvol JC, et al. Differences in MDS-UPDRS scores based on Hoehn and Yahr stage and disease duration. *Movement Disorders Clinical Practice* 2017;**4**(4):536-44. [DOI: 10.1002/mdc3.12476]

186. Labeit B, Claus I, Muhle P, Regner L, Suntrup-Krueger S, Dziewas R, Warnecke T. Effect of cognitive and motor dual-task on oropharyngeal swallowing in Parkinson's disease. *European Journal of Neurology* 2021;**28**(3):754-762.

187. Battista P, Aresta S, Tagliente S, Merlo F, Mongelli D. Exploring the neuropsychological correlates of swallowing disorders in people with Parkinson's disease: a cross-sectional study. *Dysphagia* 2025;**40**(1):292-304.

188. Cichero JA, Lam P, Steele CM, Hanson B, Chen J, Dantas RO, et al. Development of international terminology and definitions for texture-modified foods and thickened fluids used in dysphagia management: the IDDSI framework. *Dysphagia* 2017;**32**(2):293-314. [DOI: 10.1007/s00455-016-9758-y.]

189. Hirschwald J, Duncan S, Warnecke T, Boyle G, Regan J, Walshe M. Development of a Core Outcome Set for Dysphagia Interventions in Parkinson's disease (COS-DIP): study protocol. *BMJ Open* 2024;**14**(2):e076350.

190. Cattaneo C, Flynn E, Walshe M. Treatment fidelity in behavioural interventions for oropharyngeal dysphagia in Parkinson's disease. *Dysphagia* 2020;**35**(1):1443.

191. Hirschwald J, Finnegan L, Hofacker J, Walshe M. Underserved groups in dysphagia intervention trials in Parkinson's disease: a scoping review. *Ageing Research Reviews* 2024;**93**:102150.

192. Labeit B, Berkovich E, Claus I, Roderigo M, Schwake AL. Dysphagia for medication in Parkinson's disease. *npj Parkinson's Disease* 2022;**8**(1):156.

193. Ernst M, Folkerts AK, Gollan R, Lieker E, Caro-Valenzuela J, et al. Physical exercise for people with Parkinson's disease: a systematic review and network meta-analysis. *Cochrane Database of Systematic Reviews* 2024, Issue 4. Art. No: CD013856. [DOI: 10.1002/14651858.CD013856.pub3]

194. Höglinger G, Trenkwalder C; German Parkinson's Guidelines Committee. Diagnosis and treatment of Parkinson's disease (guideline of the German Society for Neurology). *Neurological Research and Practice* 2024;**6**(1):30. [DOI: 10.1186/s42466-024-00325-4]

195. Hopewell S, Chan A, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. CONSORT 2025 explanation and elaboration: updated guideline for reporting randomised trials. *BMJ* 2025;**389**:e081124. [PMID: 40228832]

196. Hoffmann TC, Glasziou PP, Boutron I, Milne R, Perera R, Moher D, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ* 2014;**348**:g1687. [PMID: 24609605]

ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis table, by intervention type

All but one included study were parallel-group RCTs; Xie 2018 used a cross-over design.

All participants were adults (18+ years) with Parkinson's disease.

Study name (year) country	Intervention details	Sample size	Outcome domains	Outcome measures	Measurement time point(s)	Synthesis method
Intervention category: behavioural intervention						
Claus 2021 Germany	EMST	45	1. Swallowing efficiency 2. Swallowing safety 3. Dysphagia severity 4. Quality of life	1. Bolus Residue Scale 2. PAS 3. SDQ 4. SWAL-QoL	4 weeks (end of treatment) 3 months (follow-up)	1. EE 2. MA 3. EE 4. EE
Feng 2019 China	Vocalisation training+ SLT	60	1. Dysphagia severity 2. Saliva management	1. SSA 2. Drooling severity (DSFS-S), drooling frequency (DSFS-F)	4 weeks (end of treatment)	1. EE 2. EE
Khan 2025 Pakistan	Biofeedback for tongue	12	Dysphagia severity	FOIS	12 weeks (end of treatment)	Summary
Logemann 2008 USA	Compensatory manoeuvres Diet modifications	NR	Swallowing safety	PAS	3 months (end of treatment)	Summary
Manor 2013 Israel	VAST+SLT	42	1. Swallowing efficiency 2. Swallowing safety 3. Dysphagia severity 4. Quality of life	1. Bolus Residue Scale 2. PAS 3. SDQ 4. SWAL-QoL	4 weeks (end of treatment) 6 months (follow-up)	1. EE 2. EE 3. EE 4. EE
Mohseni 2023 Iran	Music therapy+SLT	33	Dysphagia severity	SDQ	4 weeks (end of treatment) 3 months (follow-up)	EE
Plaza 2022 Chile	Tongue strength	60	1. Dysphagia severity 2. Quality of life	1. FOIS 2. SWAL-QoL	8 weeks (end of treatment)	1. EE 2. EE

Table 1. Overview of included studies and synthesis table, by intervention type *(Continued)*

	train-ing+SLT					
Riboldazzi 2020 Italy	EFA+SLT	25	1. Quality of life 2. Respiratory outcomes 3. Adverse events	1. PDQ-39 2. Peak expiratory flow 3. Number of hospitalisations	12 months (end of treatment)	1. EE 2. EE
Troche 2010 USA	EMST	68	1. Swallowing safety 2. Quality of life	1. PAS 2. SWAL-QoL	4 weeks (end of treatment)	1. EE 2. MA
Troche 2023 USA	EMST	65	Respiratory outcomes	Peak expiratory flow	5 weeks (end of treatment)	EE
Intervention category: neuromuscular electrical stimulation						
Baijens 2013 Netherlands	NMES	109	1. Swallowing efficiency 2. Swallowing safety	1. Bolus Residue Scale 2. PAS	4 weeks (end of treatment)	Summary
Heijnen 2012 Netherlands	NMES	109	1. Dysphagia severity 2. Quality of life	1. FOIS 2. SWAL-QoL	4 weeks (end of treatment)	Summary
Park 2018 Republic of Korea	NMES+SLT	18	1. Swallowing efficiency 2. Swallowing safety	1. VDS pharyngeal phase 2. PAS	4 weeks (end of treatment)	EE
Intervention category: transcranial direct current stimulation						
Dashtelei 2024 Iran	tDCS	29	1. Swallowing safety 2. Dysphagia severity	1. PAS 2. SDQ	2 weeks (end of treatment) 1 month (follow-up)	1. EE 2. EE
Intervention category: repetitive transcranial magnetic stimulation						
Khedr 2019 Egypt	rTMS	33	1. Swallowing efficiency 2. Swallowing safety 3. Dysphagia severity	1. Bolus Residue Scale 2. PAS 3. SDQ	2 weeks (end of treatment)	1. EE 2. EE 3. EE
Zhang 2025 China	rTMS vs NMES	113	1. Dysphagia Severity 2. Saliva management	1. FOIS 2. RSST	20 days (end of treatment)	1. EE 2. EE
Intervention category: deep brain stimulation						
Cebi 2025 Germany	DBS	20	1. Swallowing safety 2. Dysphagia severity	1. PAS 2. FOIS	8 weeks (end of treatment)	Summary

Table 1. Overview of included studies and synthesis table, by intervention type *(Continued)*

			3. Quality of life	3. SWAL-QoL		
Xie 2018 USA	DBS	15	1.Swallowing safety 2. Dysphagia severity	1. PAS 2. SDQ	14.5 months (follow-up)	Summary

DBS: deep brain stimulation; **DSFS**: Drooling Severity and Frequency Scale; **EE**: effect estimate; **EFA**: expiratory flow acceleration; **EMST**: expiratory muscle strength training; **FOIS**: Functional Oral Intake Scale; **MA**: meta-analysis; **NMES**: neuromuscular electrical stimulation; **NR:** not reported; **PAS**: Penetration–Aspiration Scale; **PDQ-39**: Parkinson's Disease Questionnaire 39; **RCT**: randomised controlled trial; **RSST**: Repetitive Saliva Swallowing Test; **rTMS**: repetitive transcranial magnetic stimulation; **SDQ**: Swallowing Disturbance Questionnaire; **SLT**: speech-language therapy; **SSA**: Standardised Swallowing Assessment; **SWAL-QoL**: Swallowing Quality of Life; **tDCS**: transcranial direct current stimulation; **UPDRS**: Unified Parkinson's Disease Rating Scale; **VAST**: video-assisted swallowing therapy; **VDS:** Videofluoroscopic Dysphagia Scale

Table 2. Studies included in qualitative synthesis

The following studies were not included in the review's quantitative analysis as they reported data in forms unsuitable for analysis.

Study	Intervention 1	Intervention 2	Intervention 3
Baijens 2013	NMES – Motor level + SLT	NMES – Sensory level + SLT	SLT
Cebi 2025	STN + SNr stimulation	STN stimulation	—
Heijnen 2012	NMES – Motor level + SLT	NMES – Sensory level + SLT	SLT
Khan 2025	Biofeedback training with 'tongueometer'	Traditional isometric tongue strengthening exercises	—
Logemann 2008	Compensatory manoeuvres	Diet modification	Diet modification
Xie 2018	DBS – 130 Hz stimulation	DBS – 60 Hz stimulation of the STN	Sham DBS

DBS: deep brain stimulation; **NMES**: neuromuscular electrical stimulation; **SLT**: speech-language therapy; **SNr**: substantia nigra pars reticulata; **STN**: subthalamic nucleus

INDEX TERMS

Medical Subject Headings (MeSH)

Bias; Deglutition; *Deglutition Disorders [etiology] [rehabilitation]; Humans; *Parkinson Disease [complications]; Quality of Life; Randomized Controlled Trials as Topic